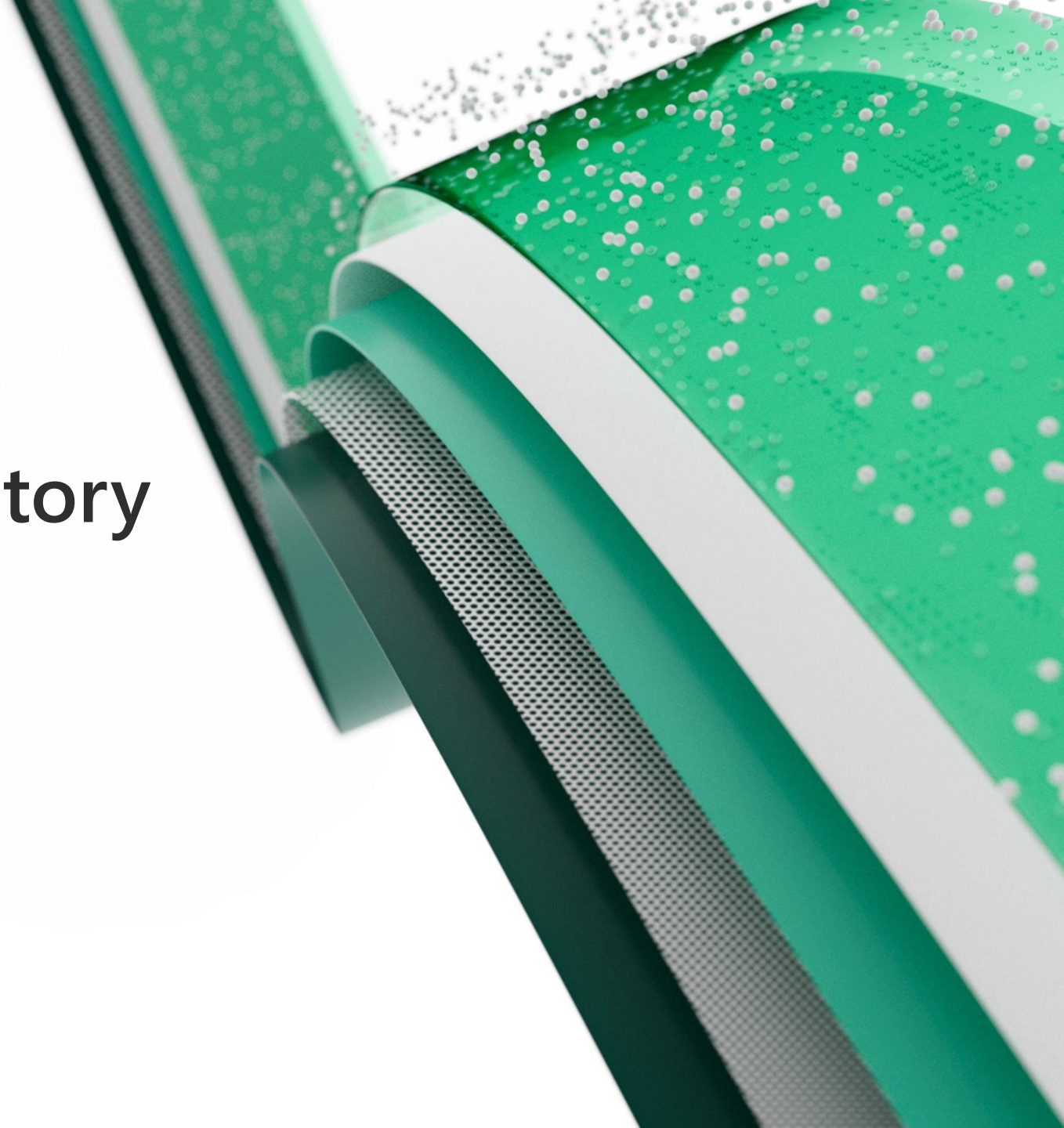
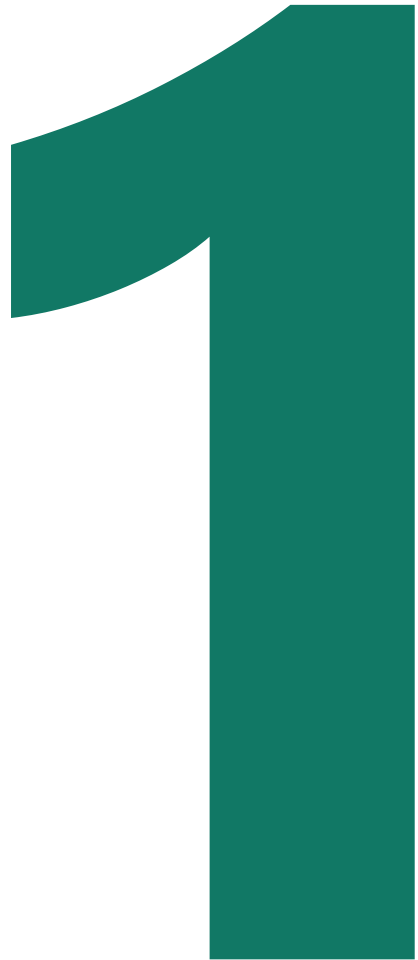


Microsoft Fabric Data Factory

Fabric Data Factory Overview





Data Factory in Microsoft Fabric



Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Pipelines



Connectors



Dataflows

Fabric Platform



AI

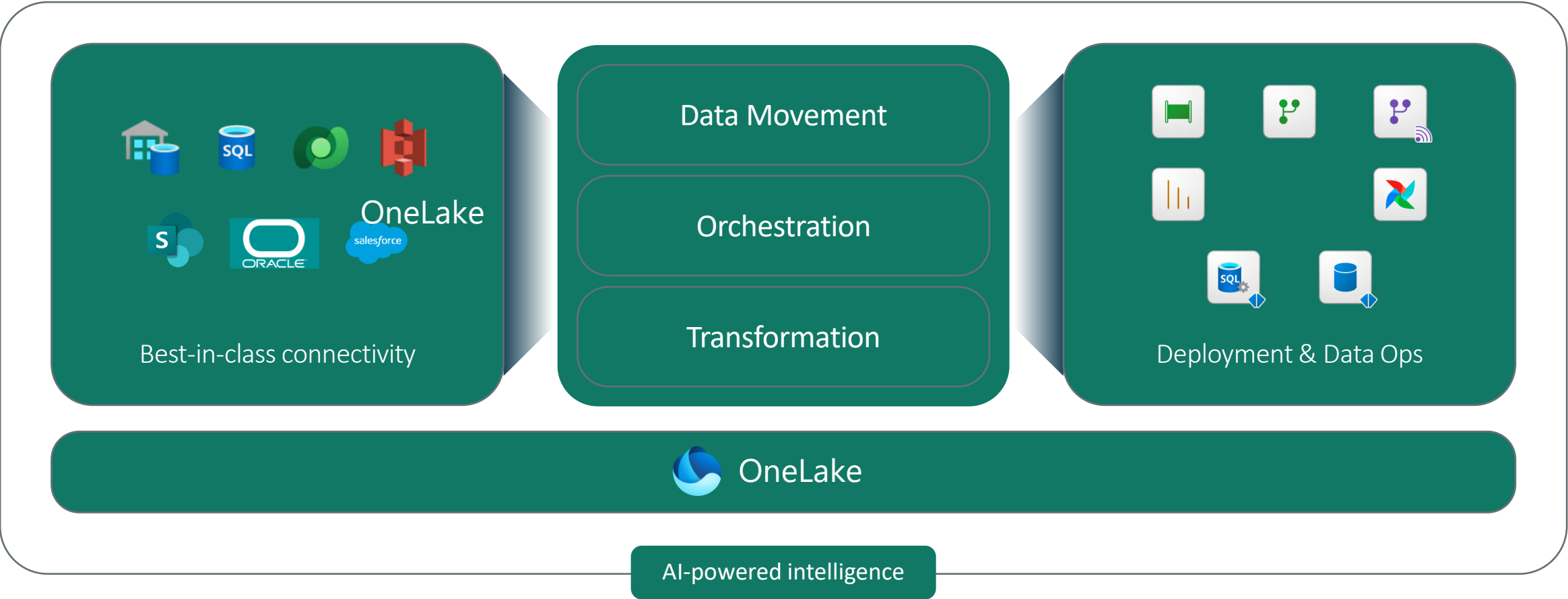


OneLake



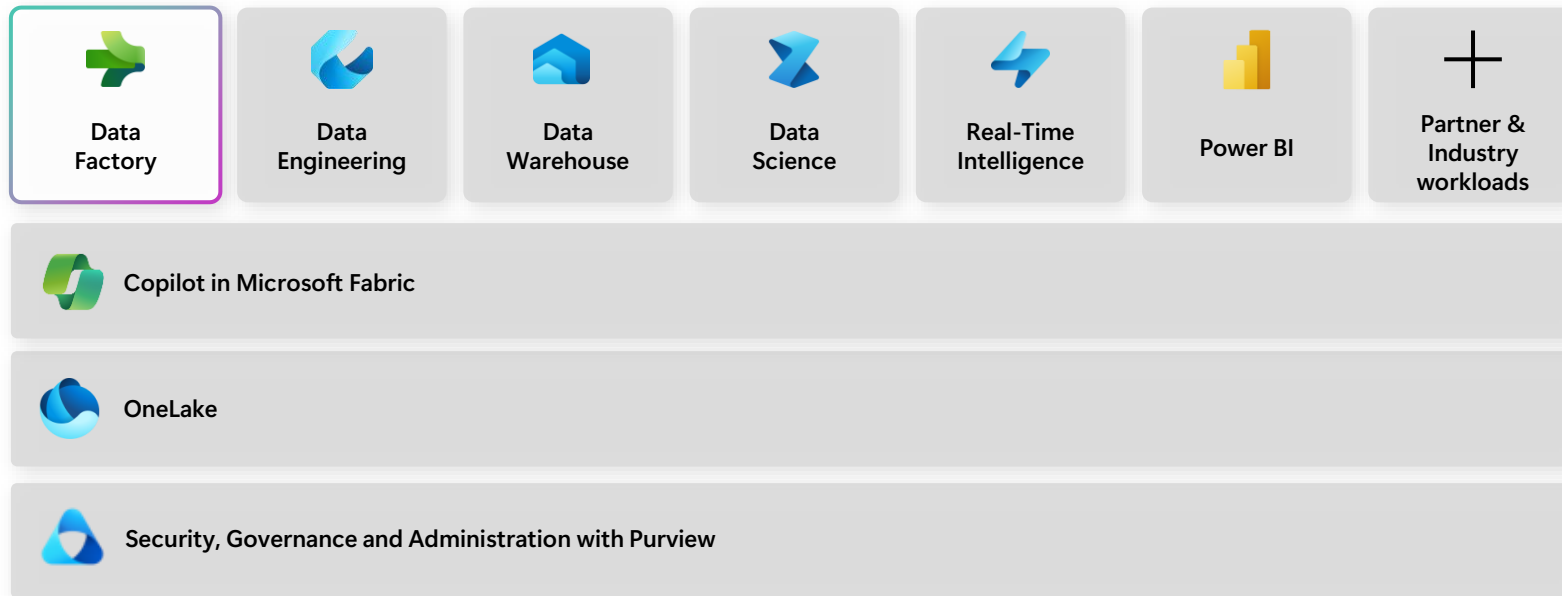
Governance

Data Factory - Microsoft Fabric



Data Factory workload

Dataflows and data pipelines bring together low-code, AI-based experiences, multi-cloud connectivity, and persistent data security and governance to help solve complex ETL scenarios for all developers



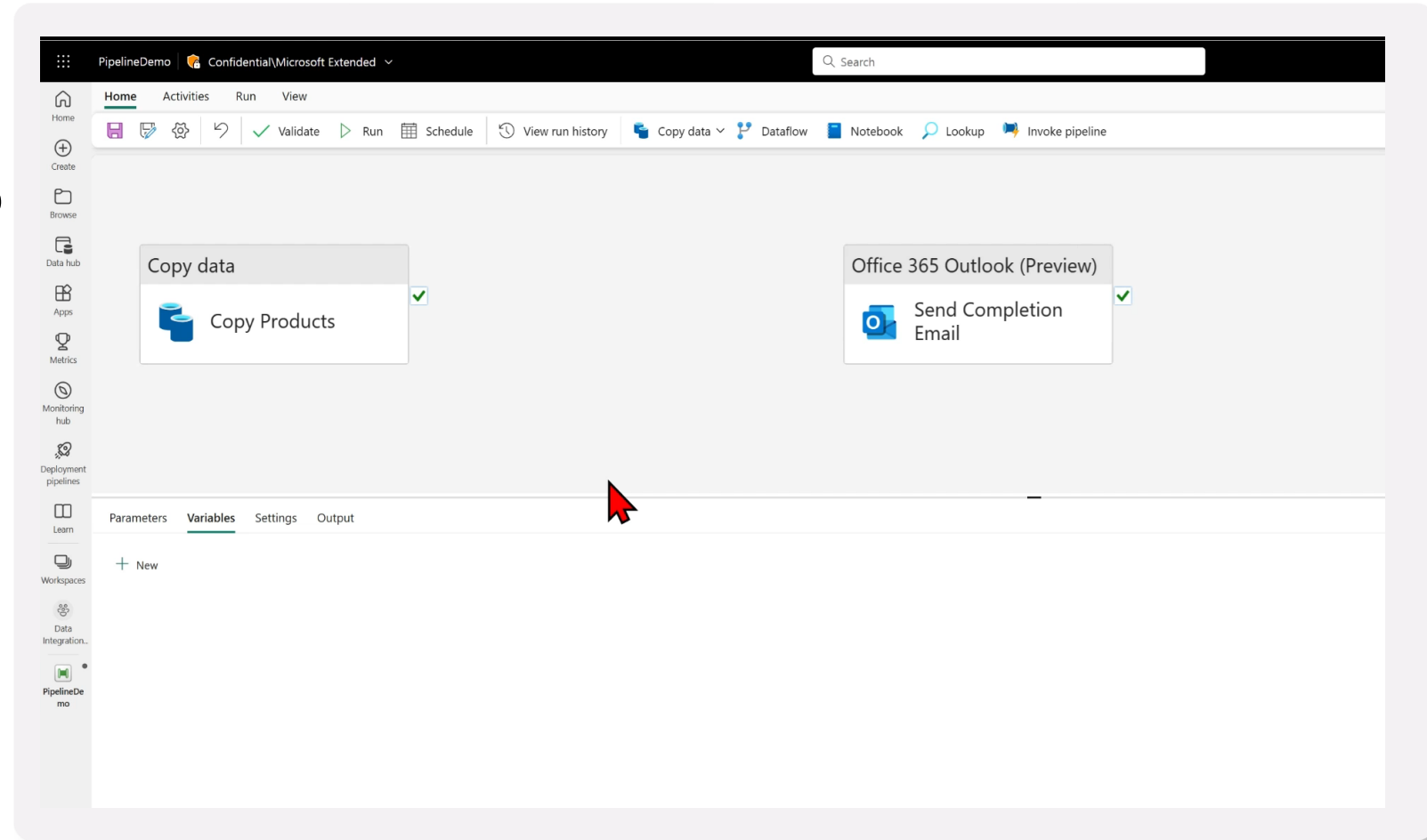
- > 200+ native data source connectors
- > 300+ data transformations in dataflows designer to transform data more easily
- > Cloud-scale data movement with Data Factory
- > Low-code interface for ingesting data from hundreds of data sources using Dataflows Gen2
- > Out-of-the-box rich data orchestration capabilities to compose flexible workflows
- > Powerful, enterprise-grade Data Factory workload with the best of ADF and Power Query together

Data Integration for everyone

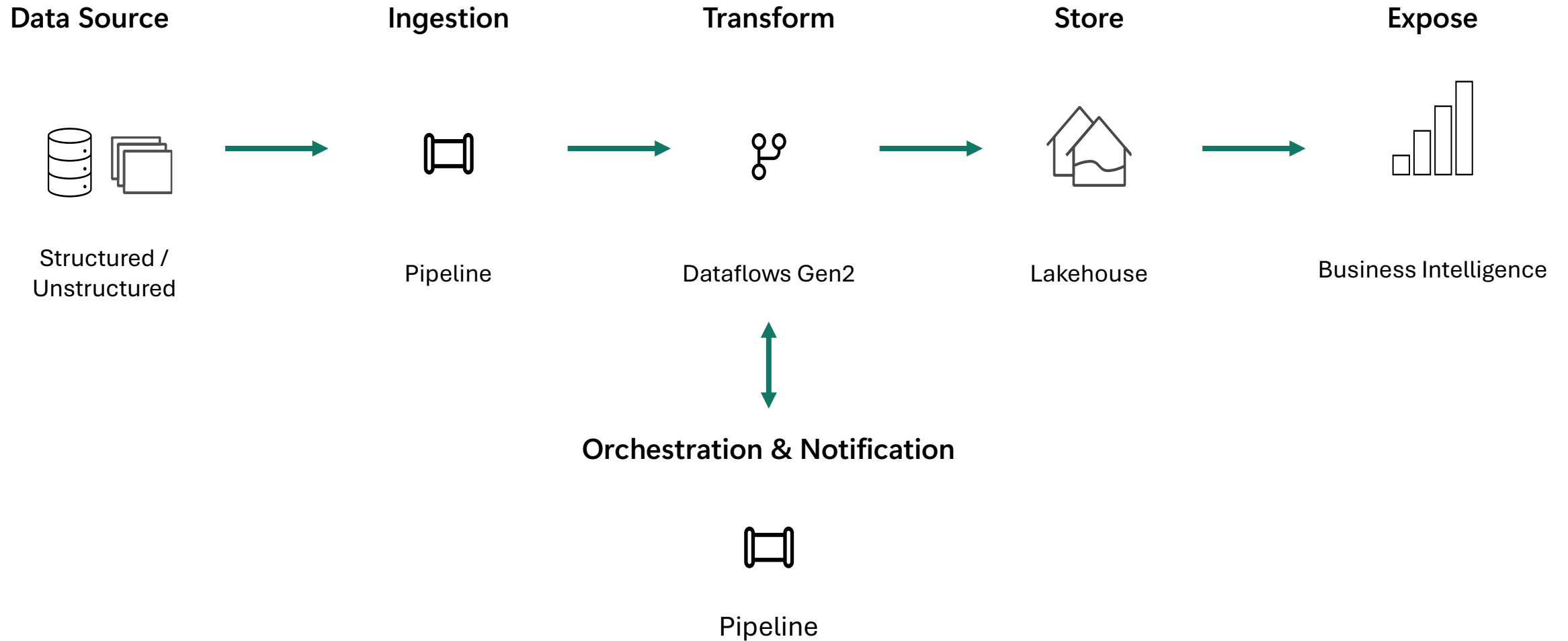
Empowering **every person** to integrate data

From **citizen** to **professional** developers

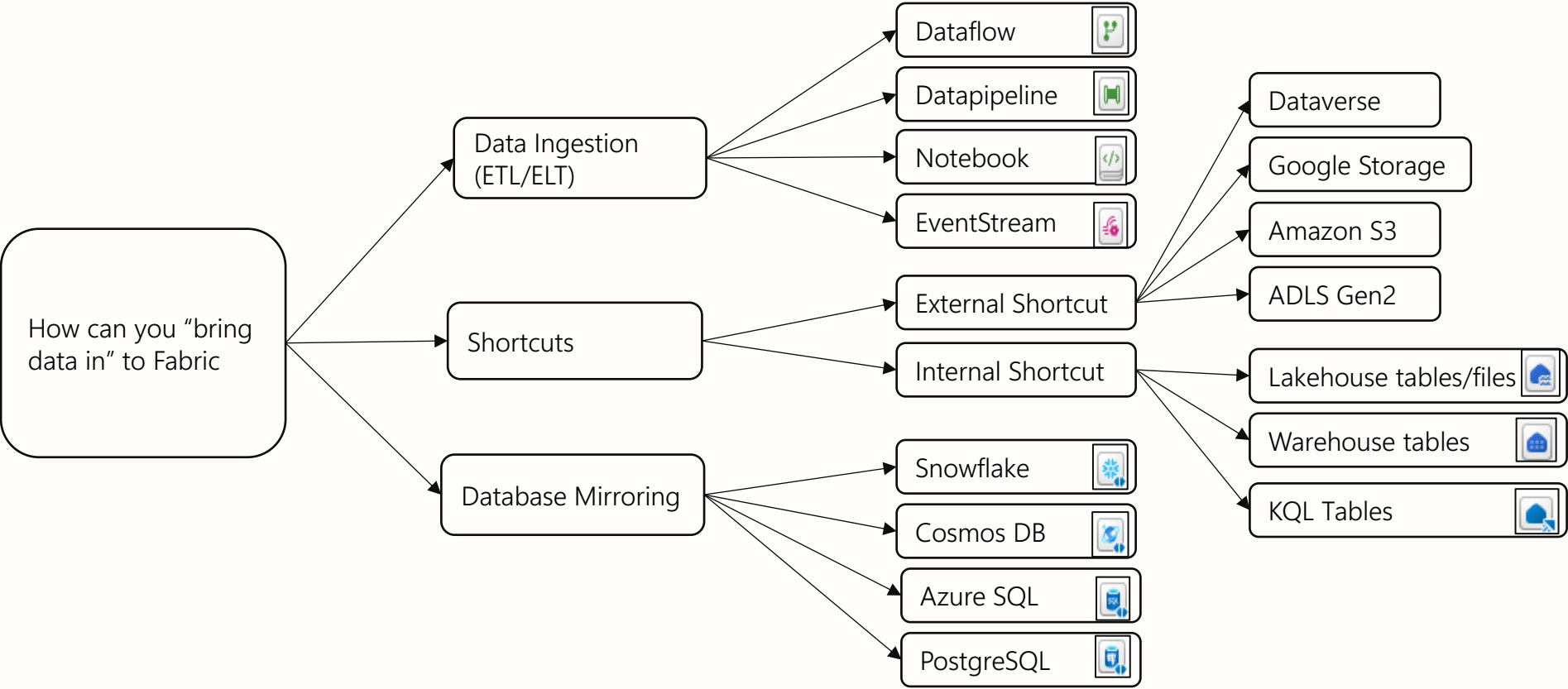
Enterprise-scale data ingestion, transformation, orchestration with **Dataflows Gen2** and **Data pipelines**



Data Factory scenario



Bring your data into Fabric

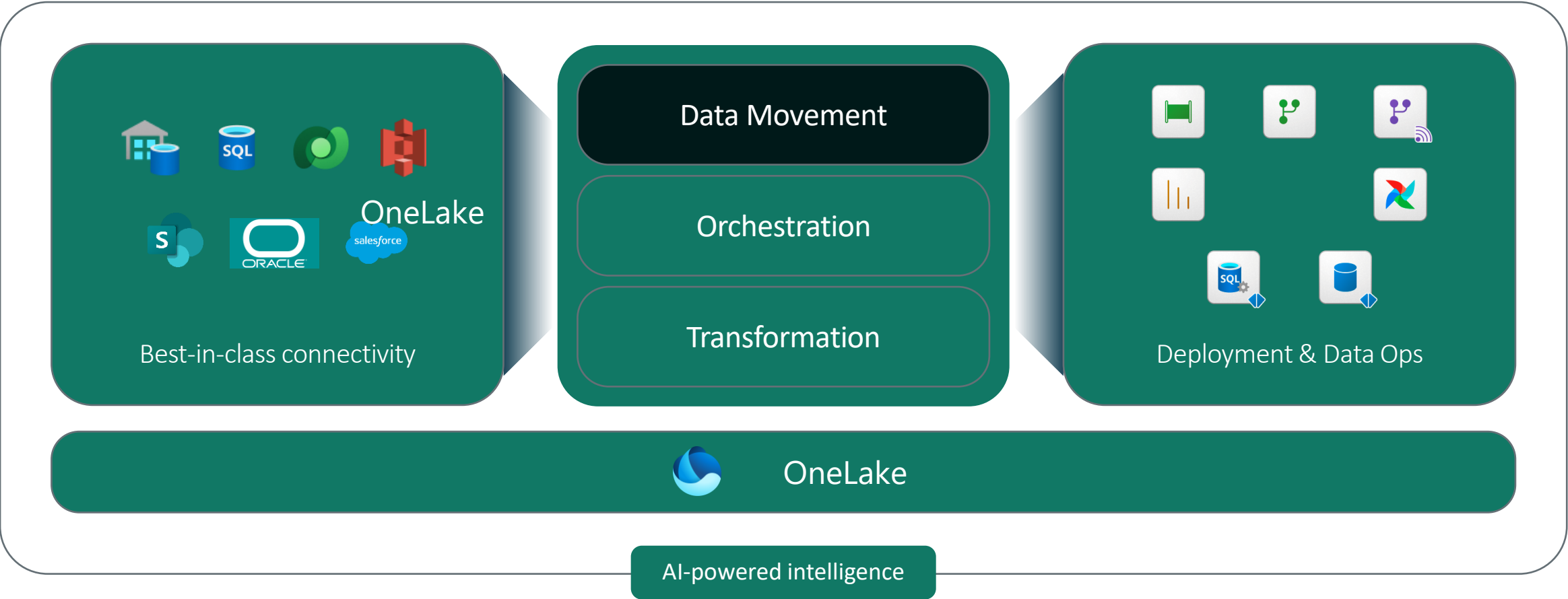


2

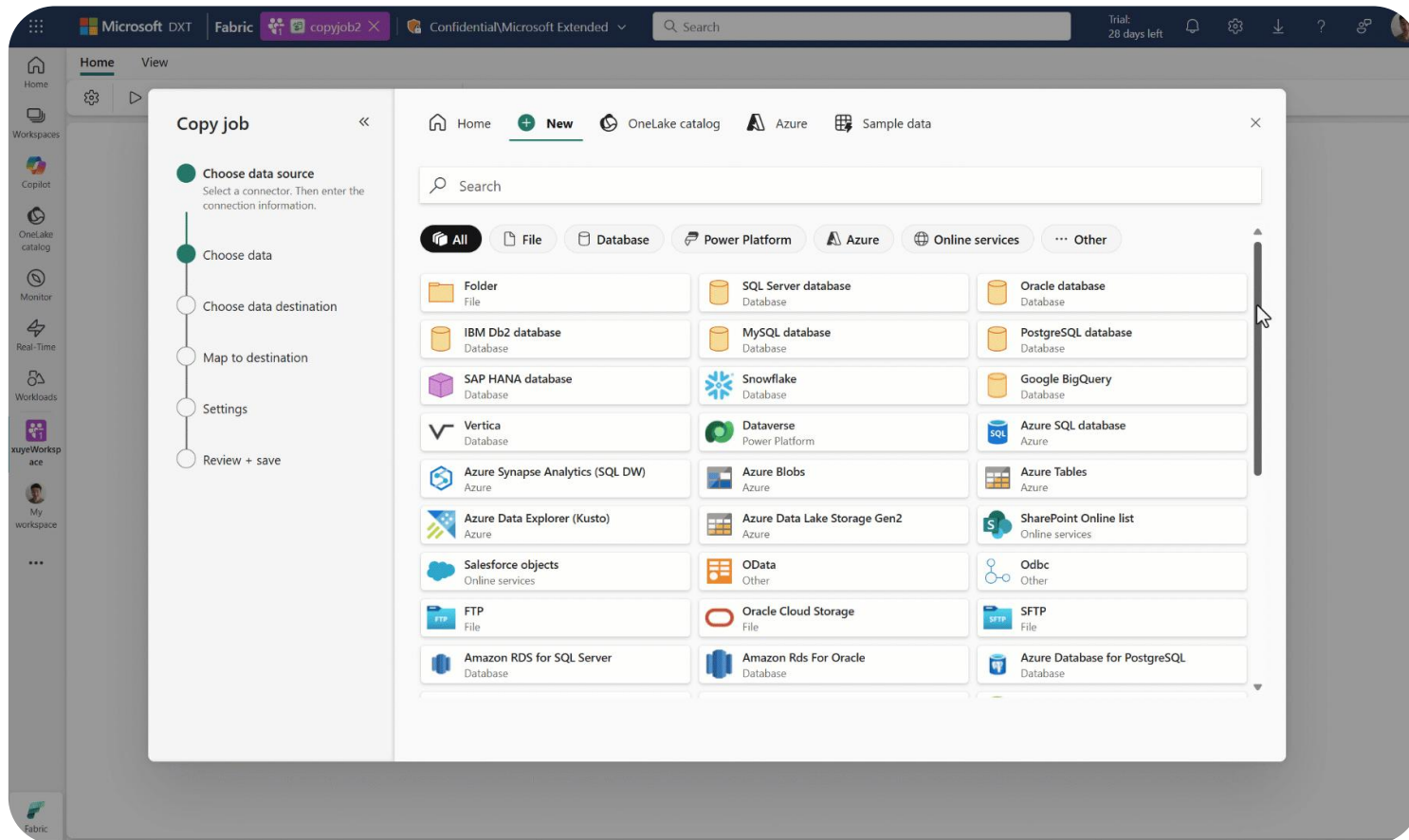
Data Pipelines



Data Factory – Data Movement



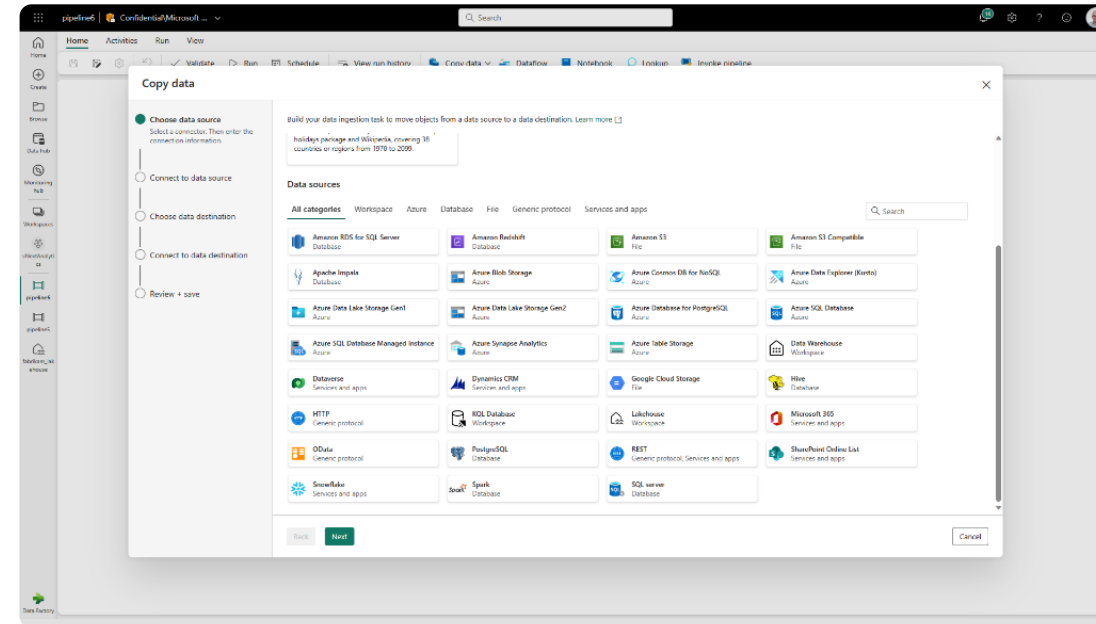
Data Factory – Copy Activity



- Copy Activity enables petabyte scale data movement in pipelines
- Copy job simplifies data movement, with built-in batch and incremental copy including change data capture replication patterns
- File, Relational, NoSQL, protocol, SaaS and Azure sources & destinations
- Connect to on-premises & cloud sources with network security compliant access to secured resources

Data pipelines

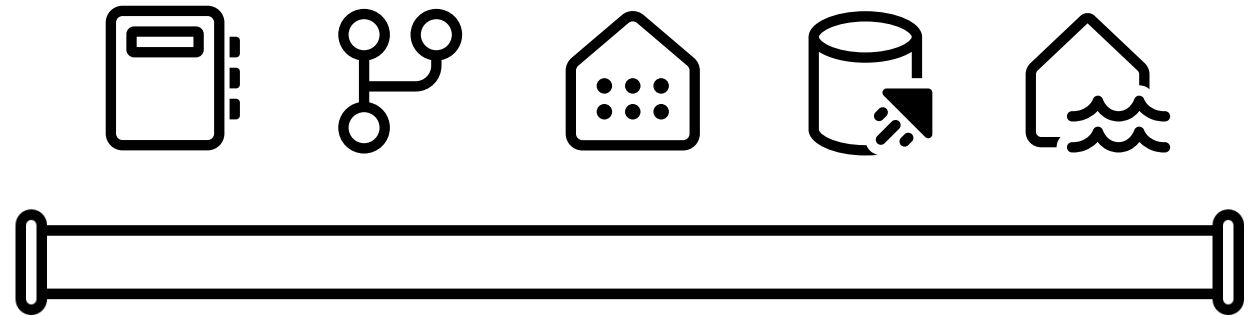
- Pipelines are code-free, serverless sequence of activities that orchestrate an overall process
- Usually extracts data from one or more sources and loads into a destination
- Commonly used to automate extract (E) from operational store
- Transform, and load (TL) into an analytical store
- Works at cloud-scale to build complex workflows
- Consider pipelines to move PB-size data, and defining sophisticated control flow pipelines



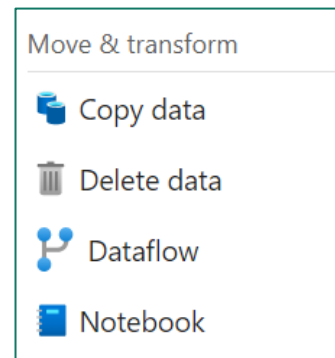
Data Pipelines continued

Pipelines come with built-in integration to other Fabric Experiences.

1. Notebooks
2. Dataflows
3. Warehouse
4. KQL Database
5. Lakehouse (Files/Tables)



Common activities



Core Concepts

- Pipelines
 - Core orchestration unit in Data Factory
 - Define workflow logic for data movement and transformation
 - Can include multiple activities chained together for complex ETL processes
- Connections
 - Enable secure access to data sources and destinations
 - Support a wide range of connectors (SQL, ADLS Gen2, Lakehouse, SaaS apps)
 - Authentication options include OAuth, Service Principal, and Managed Identity
- Activities
 - Building blocks inside pipelines (more on the next slide)

Core Concepts continued

- Activities: Executable tasks in a pipeline
- Create a flow of activities by connecting them in a sequence
- The outcome of a particular activity can be used to connect to the next activity in the sequence.
- There are two broad categories of activity in a pipeline.
 - Data transformation activities: Those that encapsulate data transfer operations (e.g., copy activity)
 - Control flow activities: Those that can implement loops, conditional branching, or manage variable and parameter values
- Parameters: Enables to provide specific values to be used each time a pipeline is run.
 - Increases the reusability of your pipelines
- Pipeline runs: Either on-demand or scheduled

Activities

Build a data pipeline to organize and move your data

Start with a blank canvas



Pipeline activity

Automate data orchestrations using rich no-code activities.

Start with guidance



Copy data assistant

Follow guided steps to copy data into Microsoft Fabric, as well as other data stores.



Practice with sample data

Quickly build a data pipeline with a predefined template to load data into Lakehouse.



Templates

Generate a new data pipeline quickly using a predefined data scenario.

Need help? [Watch a demo](#)

Move and transform

Copy data

Dataflow

Delete data

Metadata and validation

Lookup

Get Metadata

Control flow

If Condition

Switch

Filter

Wait

ForEach

Until

Set variable

Append variable

Fail

Orchestrate

Invoke Pipeline (Legacy)

Invoke Pipeline (Preview)

Web

WebHook

Semantic model refresh (Preview)

Azure Databricks

Functions

Azure HDInsight

Azure Batch

Notifications

Office 365 Email

Office 365 Outlook (Legacy)

Teams (Preview)

Transform

Spark Job Definition

Notebook

Script

Stored procedure

KQL

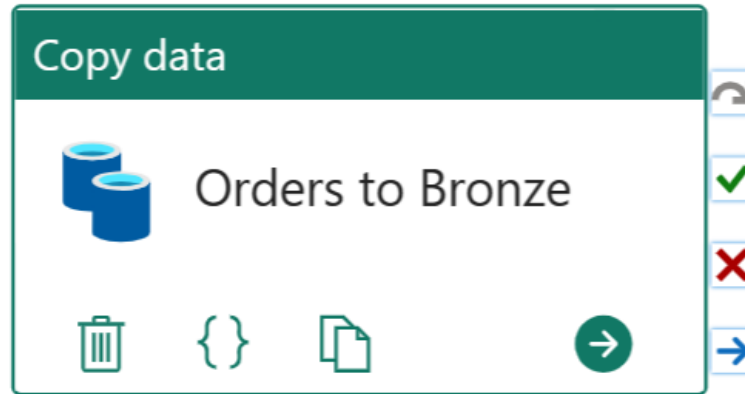
Machine Learning

Azure Machine Learning

Copy Data Activity

The Copy Activity is the **core** of Data Pipelines. In Fabric, this activity can ingest data from external systems directly into Fabric items, across Fabric items, and even push data out to external systems.

** Copy Activity can only interact with Fabric Items within the same workspace as the Data Pipeline; must create a Shortcut to interact with data in other workspaces*



Available today

45 Source Connections

32 Destination Connections

Workspace Source and Destination include

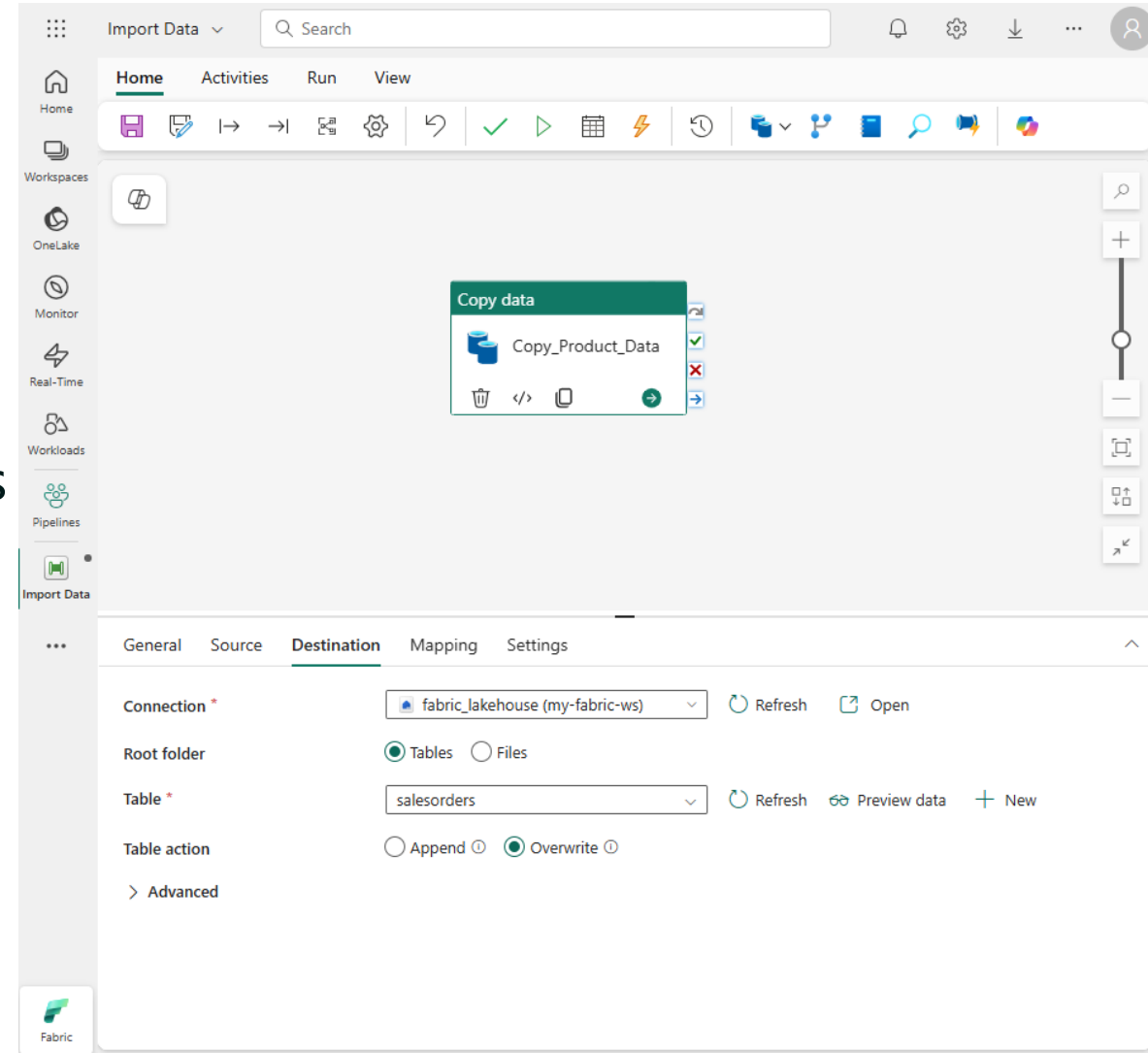
- **Lakehouse** (Tables & Files)
- **Warehouse**
- **KQL Database**

The Copy Activity allows for control over:

1. File type conversion
2. Column mapping
3. Merging/Splitting of files
4. and much more..

Copy Data Activity continued

- Can start with it on a clean canvas
- Is the most common uses of a data pipeline
- Normally combined with other activities to create a repeatable ingestion process
- After adding it to the canvas, edit its settings in the pane underneath to setup source, destination and other settings.
- When to use:
 - When you need to copy data directly between a supported source and destination without applying any transformations
 - When you want to import the raw data and apply transformations in later pipeline activities



Copy Data Assistant

- Copy Data Assistant will provide a wizard-based approach to setup the copy activity
- Configure the source, destination
Create connections as required
- Once finished, the copy activity will then be added to your pipeline canvas.
- All settings are available under the tabs when it's selected.

Copy job <<

- ✓ Choose data source
- ✓ Choose data
- ✓ Choose data destination
- ✓ Choose copy job mode
- ✓ Map to destination
- Review + save**
Confirm Copy summary

Review + save ✕

Summary

→ Source

dimension_city
Parquet

Full copy

→ Destination

CopyDestination
Lakehouse

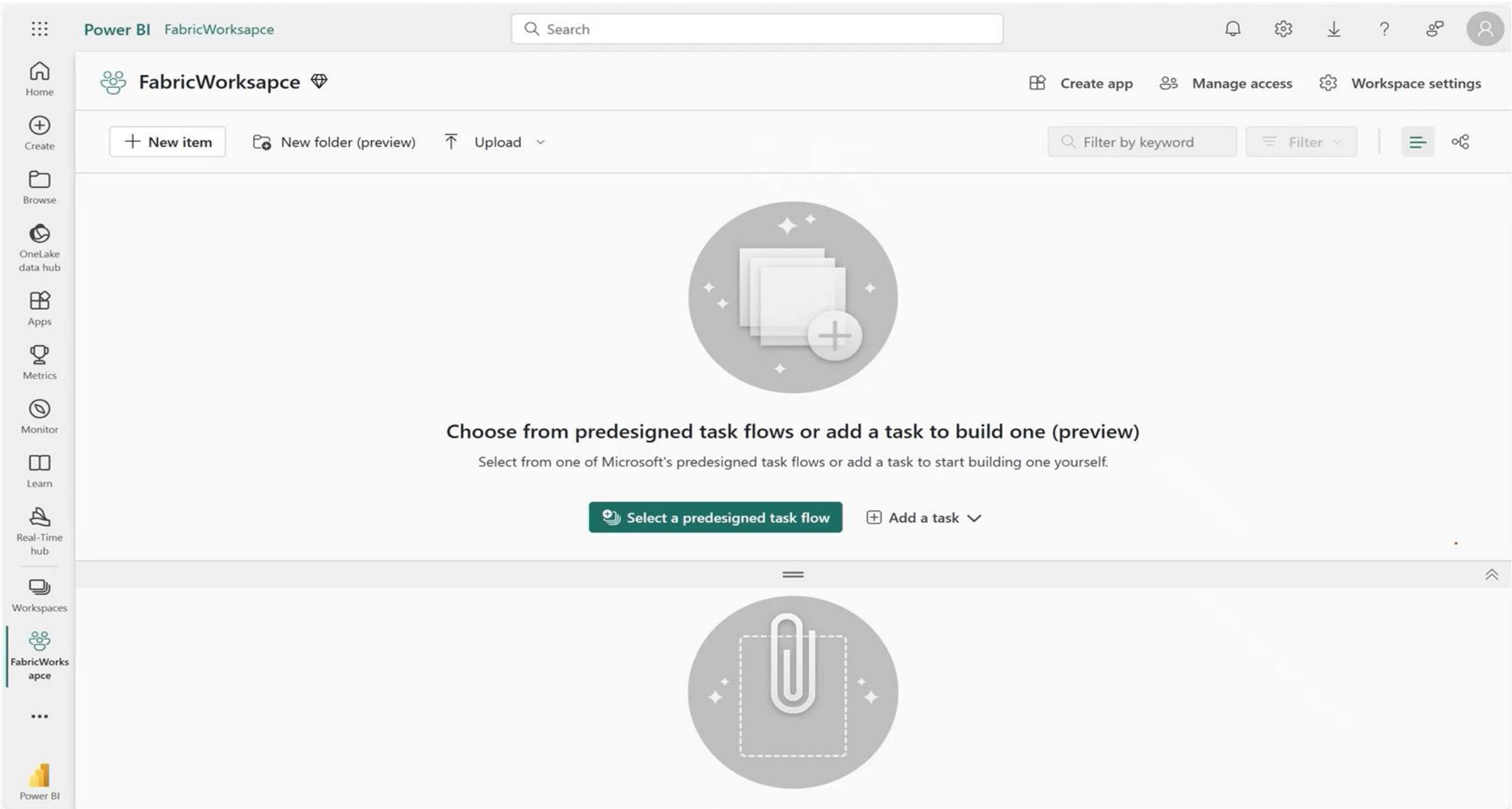
Sample dataset
dimension_city (Parquet)

Connection name
CopyDestination

Table name
sampledata_WideWorldImportersDW_parquet_full_dimension_city__parquet

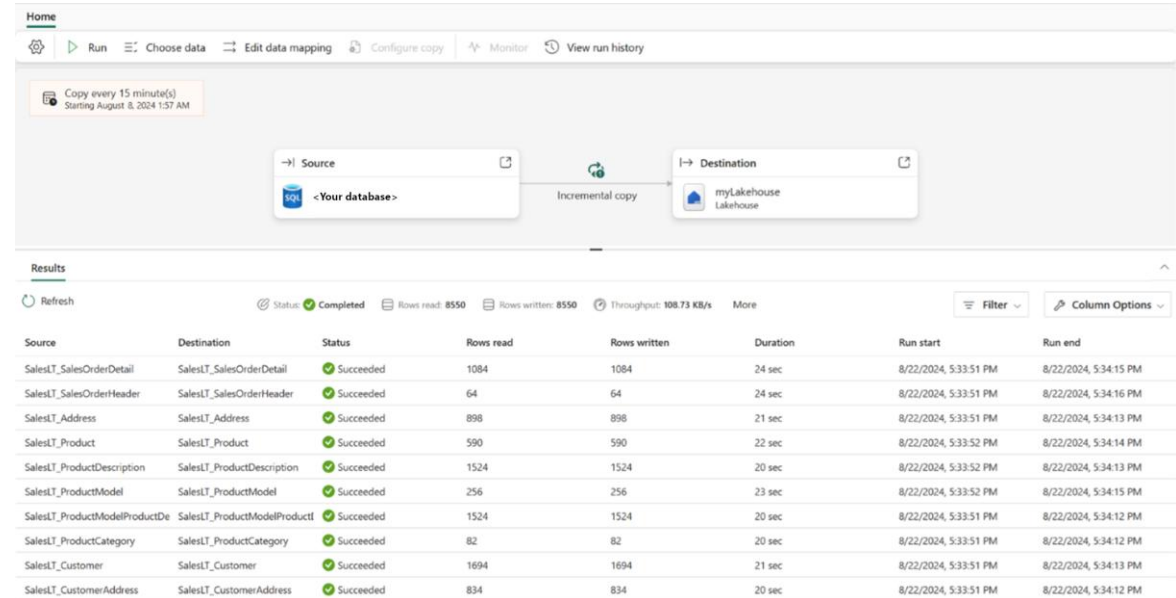
Back Save

Demo: Copy Activity



Copy job activity

- Simplifies data movement from many sources to many destinations
- No pipelines required
- Enables to manage data movement alongside transformations, notifications, and more; all in one place.
- Brings the simplicity of the Copy job item directly into your pipelines



The screenshot displays the Azure Data Factory Copy Job interface. At the top, the 'Home' tab is active, showing a summary of the job: 'Copy every 15 minute(s) Starting August 8, 2024 1:57 AM'. Below this, a visual representation of the copy job is shown, with a 'Source' box labeled '<Your database>' and a 'Destination' box labeled 'myLakehouse Lakehouse', connected by an 'Incremental copy' arrow. The 'Results' tab is selected, showing a table of job details. The table has columns for Source, Destination, Status, Rows read, Rows written, Duration, Run start, and Run end. All rows show a 'Succeeded' status. The table is filtered to show 'Completed' status, with a total of 8550 rows read and 8550 rows written, at a throughput of 108.73 KB/s.

Source	Destination	Status	Rows read	Rows written	Duration	Run start	Run end
SalesLT_SalesOrderDetail	SalesLT_SalesOrderDetail	✓ Succeeded	1084	1084	24 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:15 PM
SalesLT_SalesOrderHeader	SalesLT_SalesOrderHeader	✓ Succeeded	64	64	24 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:16 PM
SalesLT_Address	SalesLT_Address	✓ Succeeded	898	898	21 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:13 PM
SalesLT_Product	SalesLT_Product	✓ Succeeded	590	590	22 sec	8/22/2024, 5:33:52 PM	8/22/2024, 5:34:14 PM
SalesLT_ProductDescription	SalesLT_ProductDescription	✓ Succeeded	1524	1524	20 sec	8/22/2024, 5:33:52 PM	8/22/2024, 5:34:13 PM
SalesLT_ProductModel	SalesLT_ProductModel	✓ Succeeded	256	256	23 sec	8/22/2024, 5:33:52 PM	8/22/2024, 5:34:15 PM
SalesLT_ProductModelProductDe	SalesLT_ProductModelProduct	✓ Succeeded	1524	1524	20 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:12 PM
SalesLT_ProductCategory	SalesLT_ProductCategory	✓ Succeeded	82	82	20 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:12 PM
SalesLT_Customer	SalesLT_Customer	✓ Succeeded	1694	1694	21 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:13 PM
SalesLT_CustomerAddress	SalesLT_CustomerAddress	✓ Succeeded	834	834	20 sec	8/22/2024, 5:33:51 PM	8/22/2024, 5:34:12 PM

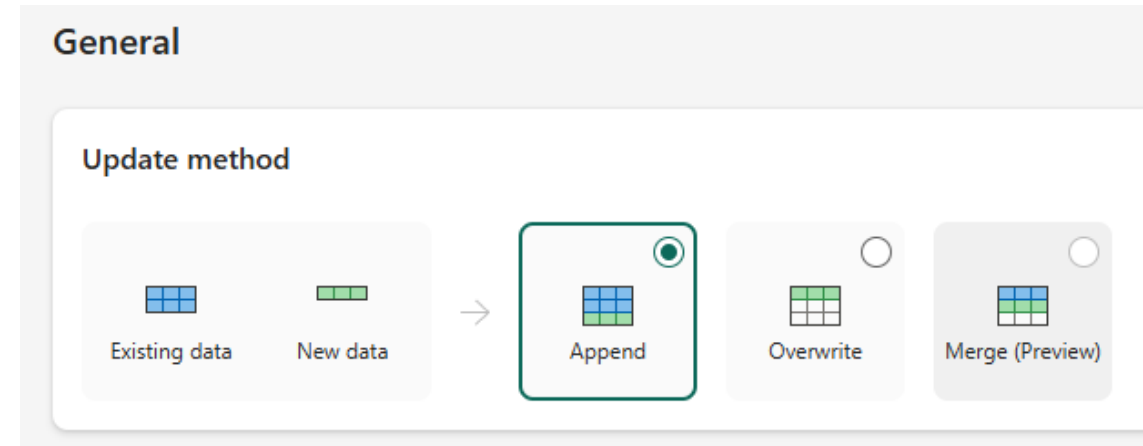
Copy Modes

- Full copy: Copies all data from source to destination
- Incremental copy: Subsequent runs after Full, copies new or changed data
 - Databases: Only new or updated rows are copied
 - Storage: Only files with a newer LastModifiedTime are copied
- If a copy job fails, the next run resumes from the state of the last successful run
- Incremental copy on supported sources (e.g., Azure SQL) can use:
 - Water-mark based incremental copy
 - CDC based incremental copy (Preview)

The screenshot shows a 'Copy job' configuration window with a sidebar on the left and a main panel on the right. The sidebar contains a vertical list of steps: 'Choose data source' (checked), 'Choose data' (checked), 'Choose data destination' (checked), 'Choose copy job mode' (active), 'Map to destination', and 'Review + save'. The main panel is titled 'Choose copy job mode' and contains a sub-header 'Choose copy job mode'. Below this is a descriptive text: 'Choose how you want the data to be copied. This copy mode will be applied every time you run the job, which can be run once or multiple times. After this new copy job item is created, you can then schedule how often the job is run and monitor statuses.' There are two options: 'Full copy' (described as 'Copy all data at once') and 'Incremental copy' (described as 'Initially copies all data, subsequent runs copy only changes'). The 'Full copy' option is selected, indicated by a green border and a green radio button. At the bottom right of the main panel are 'Back' and 'Next' buttons.

Update Method

- Copy job **appends** new data – default behavior
- Can choose to:
 - Merge: update existing rows using a key column. Copy job uses the primary key by default, if one exists
 - Overwrite: replace existing data. If a file with the same name already exists, it's replaced
- Append and overwrite are supported in Full Copy
- Append and merge are supported in Incremental Copy



Reset Incremental Copy

- Reset incremental copy to full on the next run
- Useful for dealing with data discrepancy situation
- Can be reset at job level or table level
- Changing the incremental column for an existing job, will reset incremental copy to full copy

The screenshot displays the Microsoft Data Factory (ADF) interface. At the top, the navigation bar includes the Microsoft logo, the current workspace name 'DataIngestion_CDC', and a search bar. Below the navigation bar, a toolbar contains icons for 'Run', 'Schedule', 'Choose data source', 'Edit mapping', 'Advanced settings', 'View run history', and 'Reset'. The main canvas shows a data copy job configuration with a 'Source' connector (datafactory-demo-sqlserver.database.wi... Azure SQL Database) and a 'Destination' connector (datafactory-demo-sqlserver.database.wi... Azure SQL Database), connected by an 'Incremental copy' activity. A tooltip indicates the job is scheduled to 'Copy every 15 minute(s)' starting on 'June 6, 2025 4:49 PM'. Below the canvas, the 'Results' tab is active, showing a table of job execution details.

Source → Destination	Status	Rows read	Rows written	Duration	Run start
dbo.SalesLT_Customer → dbo.SalesLT_Customer	✓ Succeeded	0	0	39 sec	8/29/2025, 3:00:41
dbo.SalesLT_Product → dbo.SalesLT_Product	✓ Succeeded	0	0	50 sec	8/29/2025, 3:00:41
dbo.SalesLT_ProductCategory → dbo.SalesLT_ProductCategory	✓ Succeeded	0	0	40 sec	8/29/2025, 3:00:41

Copy Job - Other capabilities

- Automatic table creation and truncation on destination
- Supports:
 - Manual Runs, Schedule Runs, Runs on Event Trigger
- Choose from multiple host options, including an on-premises gateway or a VNet gateway, to securely access data behind a firewall or within a VNet.
- Supports source control, CI/CD, Variable library
- Monitoring
 - Through the job panel
 - Monitoring Hub

Multi-cloud destinations

Microsoft

Fabric

My_Golden_Warehouse

Notebook 1

Search

Trial: 25 days left

18

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

MyDevelopmentWorkspace

My workspace

MyDevelopmentWorkspace

Development

New item

New folder

Import

Migrate

Add

View deployment pipeline

Create app

Manage access

Workspace settings

Source control

Filter by keyword

Filter

High-volume data ...

Low-volume data i...

Bronze data

Initial process

Silver data

Further transform

Golden data

Data visualize

Distribute Data

Medallion

Task flow details

Organize data in a lakehouse or warehouse while progressively improving its structure and quality through each layer, from Bronze to Silver to Gold, resulting in high-quality data that is easy to analyze.

Edit

9 Tasks

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorseme	Sensitivity	Included in app
	My_Golden_Warehouse	Synced	Lakehouse	—	Ye Xu	—	—	—	Confidential...	
	My_Golden_Warehouse	—	SQL analytic...	—	Ye Xu	—	—	—	Confidential...	
	Notebook 1	Synced	Notebook	—	Ye Xu	—	—	—	Confidential...	
	Notebook 2	Synced	Notebook	—	Ye Xu	—	—	—	Confidential...	
	Variables_Library	Synced	Variable libra...	High-vol...	Ye Xu	—	—	—	Confidential...	

Fabric

Development

Last synced: 9/5/2025 at 1:49 PM

3b3e203e

Templates

Pipeline templates



→ Import template

Filter by keyword

Filter



AI-Develop RAG pipeline using SQL database in Fabric

by Microsoft

Use this Retrieval Augmented Generation (RAG) pipeline template to get your data ready in SQL Database in fabric for building...



Bulk Copy from Database

by Microsoft

Use this template to copy data in bulk from a database using an external control table to store the partition list of your source tables....



Bulk Copy from Files to Database

by Microsoft

Use this template to copy data in bulk from Azure Data Lake Storage Gen2 to Azure SQL Database.



Copy data from ADLS Gen2 to Lakehouse file

by Microsoft

Use this template to copy data from ADLS Gen2 to a specified file location in your Lakehouse.



Copy data from ADLS Gen2 to Lakehouse Table

by Microsoft

Use this template to copy data from ADLS Gen2 to a specified table in your Lakehouse.



Copy data from Azure SQL DB to Lakehouse Table

by Microsoft

Use this template to copy data from your Azure SQL database to a specified table in your Lakehouse.



Copy data from sample data to Lakehouse file

by Microsoft

Use this template to copy data from sample data (NYC Taxi - Green) to Lakehouse file.



Copy data from sample data to Warehouse

by Microsoft

Use this template to copy data from sample data (Retail Data Model from Wide World Importers) to your Fabric Warehouse....



Copy multiple files containers between File Stores

by Microsoft

Use this template to leverage multiple copy activities to copy containers or folders between file based stores, where each copy...



Copy new files only by LastModifiedDate

by Microsoft

Use this template to copy new or changed files only by using LastModifiedDate.



Data anonymization with Presidio as an HTTP service

by Microsoft

This template uses Presidio as an HTTP service to anonymize PII data in text files during copy between one storage account to another....



Delete files older than 30 days

by Microsoft

Use this template to delete files that have been modified more than 30 days ago from storage stores.

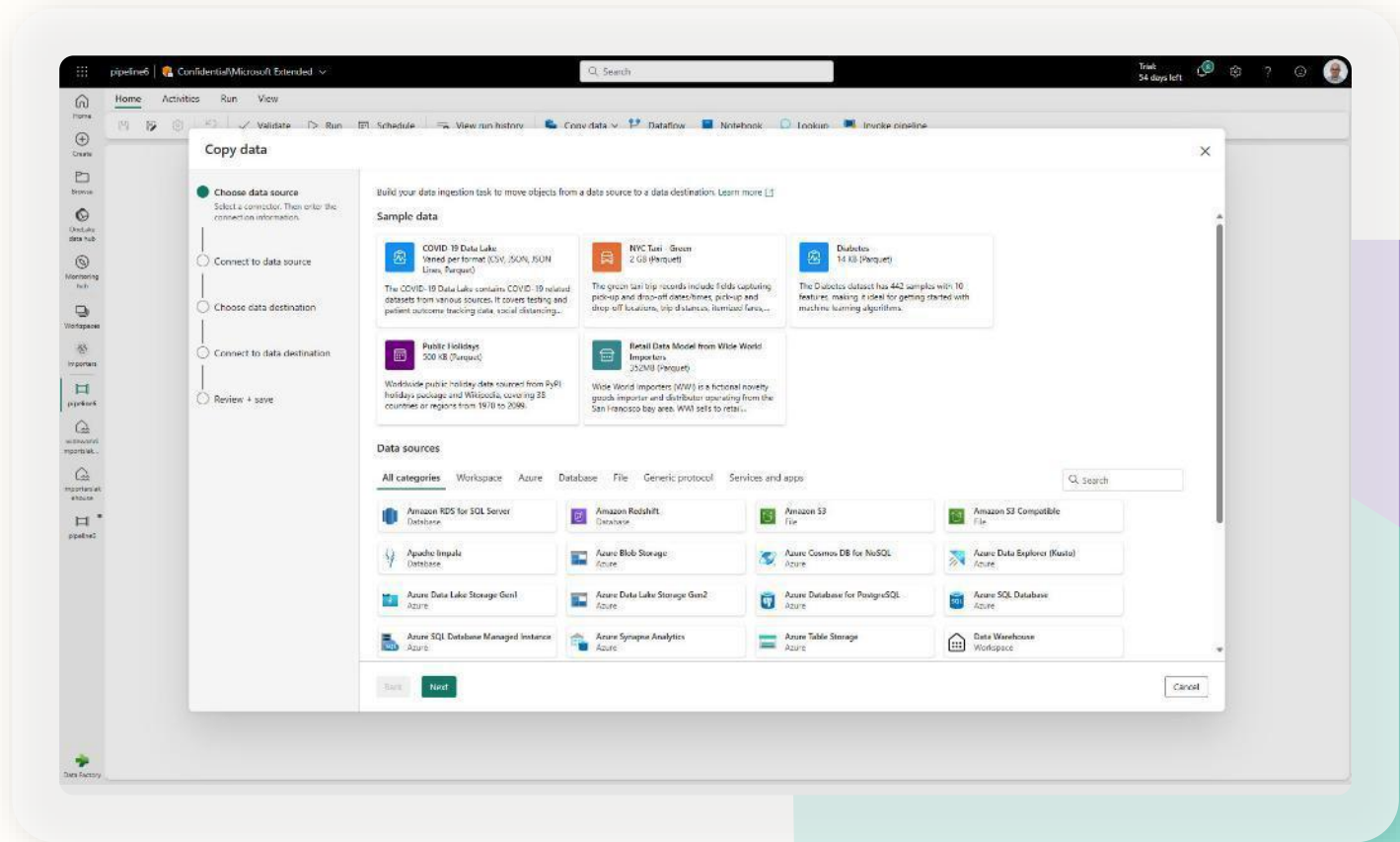
Next

Data Pipelines | Sample data

Sample Datasets helps new users get started quickly, building out their ELT processes using Data Pipelines

Sample datasets:

- COVID-19 Data Lake (CSV, JSON, JSON Lines, Parquet)
- NYC Tax – Green (2GB Parquet)
- Diabetes (14K Parquet)
- Public Holidays (500KB Parquet)
- Retail Data Model from Wide World Importers (352MB Parquet)



Pipeline as a Key Enabler

- **Data Engineering:** Automate ingestion and transformation of large datasets from multiple sources.
- **Data Science:** Prepare and deliver curated datasets for advanced modeling and experimentation.
- **Orchestration:** Coordinate complex workflows involving multiple services and dependencies.
- **Web Integration:** Connect APIs and external systems for real-time data exchange.
- **Large-Scale Copy:** Handle bulk data transfers across environments with reliability.
- **Versatility:** Support numerous use cases within Fabric, from operational analytics to enterprise BI.



For Data Engineering



For Data Science



For Orchestration



For Web Integration



For very large copy



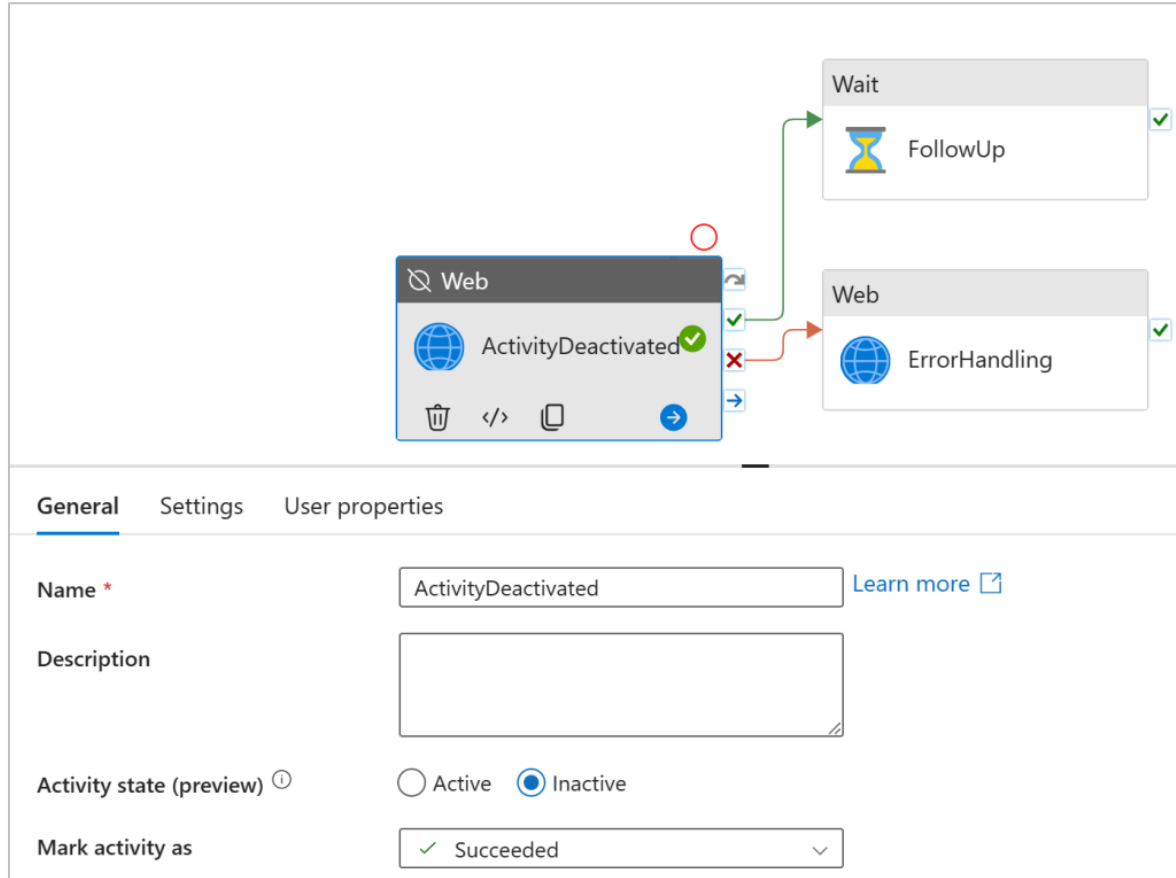
For many more use cases in Fabric!

Difference between Dataflow Gen2 and Pipelines

- Dataflows Gen2 lets you to use a low-code interface and 300+ data and AI-based transformations to you easily clean, prep, and transform data with more flexibility than any other tool.
- Data Pipelines enable rich out-of-the-box data orchestration capabilities to compose flexible data workflows that meet your enterprise needs. In a pipeline, you can create logical groupings of activities that perform a task, which might include calling a Dataflow to clean and prep your data.
- While there's some functionality overlap between the two, the choice of which to use for a specific scenario depends on whether you require the full richness of pipelines or can use the simpler but more limited capabilities of dataflows. For more information, see the [Fabric decision guide](#)

Scenario	Use Dataflow	Use Pipeline
Visual data transformation	✓	✗
Orchestrating multiple steps	✗	✓
Reusable ETL logic	✓	✓ (by calling dataflows)
Complex control flow (e.g., loops, conditions)	✗	✓
API-based automation	✗	✓

Deactivating and Reactivating Activities



The screenshot displays the Azure Logic Apps Designer interface. A pipeline is shown with three activities: 'Wait' (FollowUp), 'Web' (ErrorHandling), and 'ActivityDeactivated'. The 'ActivityDeactivated' activity is highlighted with a red circle. Below the pipeline, the 'General' tab of the activity properties is visible. The 'Name' field is set to 'ActivityDeactivated'. The 'Description' field is empty. The 'Activity state (preview)' is set to 'Inactive'. The 'Mark activity as' dropdown is set to 'Succeeded'.

General Settings User properties

Name * ActivityDeactivated [Learn more](#)

Description

Activity state (preview) ^① ☐ Active ☒ Inactive

Mark activity as ✓ Succeeded

Use Cases

- **When developing a pipeline** : use inactive copy data for iterative development process (avoid to execute all activities each time).
- **After deployment** : use inactive copy data to disable certain activities causing troubles to avoid costly retries.

... and there is so much more

Control Flow	Nesting	Parameterization	Expression Language	Repeatable
• Conditionals • Loops	• Pipelines can call other Pipelines • Bi-directional communication via output variables	• Can have many parameters of varying types	• Almost every field within <u>all</u> activities can be made dynamic! • Extensive built-in functions allow for massive customizations	• Can be scheduled to run

Pipeline Scheduling

Current Item level refresh options for datasets and **dataflows** are limited and often frustrating for customers. (3k idea votes)

With **data pipelines**, users can schedule events - by the minute, hourly, daily, or weekly and set any desired time value.

Item level refresh

The screenshot shows the 'Refresh' configuration panel. At the top, there's a 'Refresh' section with a toggle switch set to 'On'. Below it, the 'Refresh frequency' is set to 'Daily' in a dropdown menu. The 'Time' section has six time slots, each with a dropdown for the hour and a dropdown for the minute. The first slot is set to 12:00 AM. At the bottom, there's a checkbox for 'Send refresh failure notifications to' with 'Dataset owner' selected. There are 'Apply' and 'Discard' buttons at the bottom.

- Weekly/Daily and only in 30-minute increments (00, 30)
- More robust scheduling can be accomplished with REST APIs

Pipeline scheduler

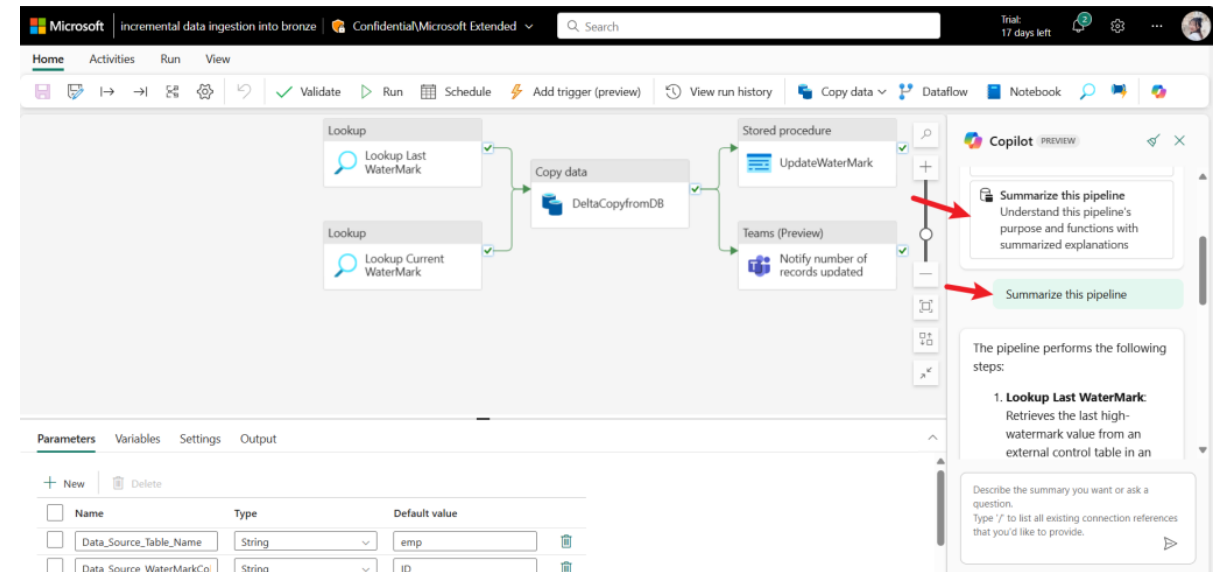
The screenshot shows the 'SalesPipeline' pipeline scheduler interface. It has a sidebar with 'Schedule' selected. The main area shows the 'Schedule' section with a 'Scheduled run' toggle set to 'On'. The 'Repeat' section has a dropdown menu set to 'Daily'. The 'Start' and 'End' fields are set to 'mm/dd/yyyy --:--:00 --'. The 'Time zone' is set to '(UTC-06:00) Central Time (US and Canada)'. There are 'Apply' and 'Run' buttons.

- Robust scheduling options and built in monitoring.
- Activities start "on time" as opposed to the current wait queue that Power BI customers are accustomed to.

Copilot in Data Factory

Easily integrate generative AI into your pipelines using Copilot

- Pipeline Generation: Describe your desired pipeline, Copilot generates the pipeline activities
- Error message assistant: Troubleshoot pipeline issues and get actionable troubleshooting guidance
- Summarize Pipeline: Explain your complex pipeline with a summary
- Build expressions: Generate and explain pipeline expressions using Copilot's



New

New items saved to :  Data Integration Copilot [Show less ^](#)



Dataflow Gen2
(Preview)



Data pipeline
(Preview)

Recommended

< >

You frequently open this



My workspace

You frequently open this



Trident-DI-Copilot

You frequently open this



Data Integration Copilot

You frequently open this



Pipeline Copilot

 Introduction 1



Watch video

Quick access

-  Recent
-  Favorites
-  Shared with me

 Filter by keyword

 Filter 

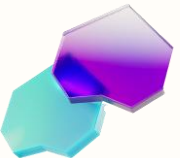
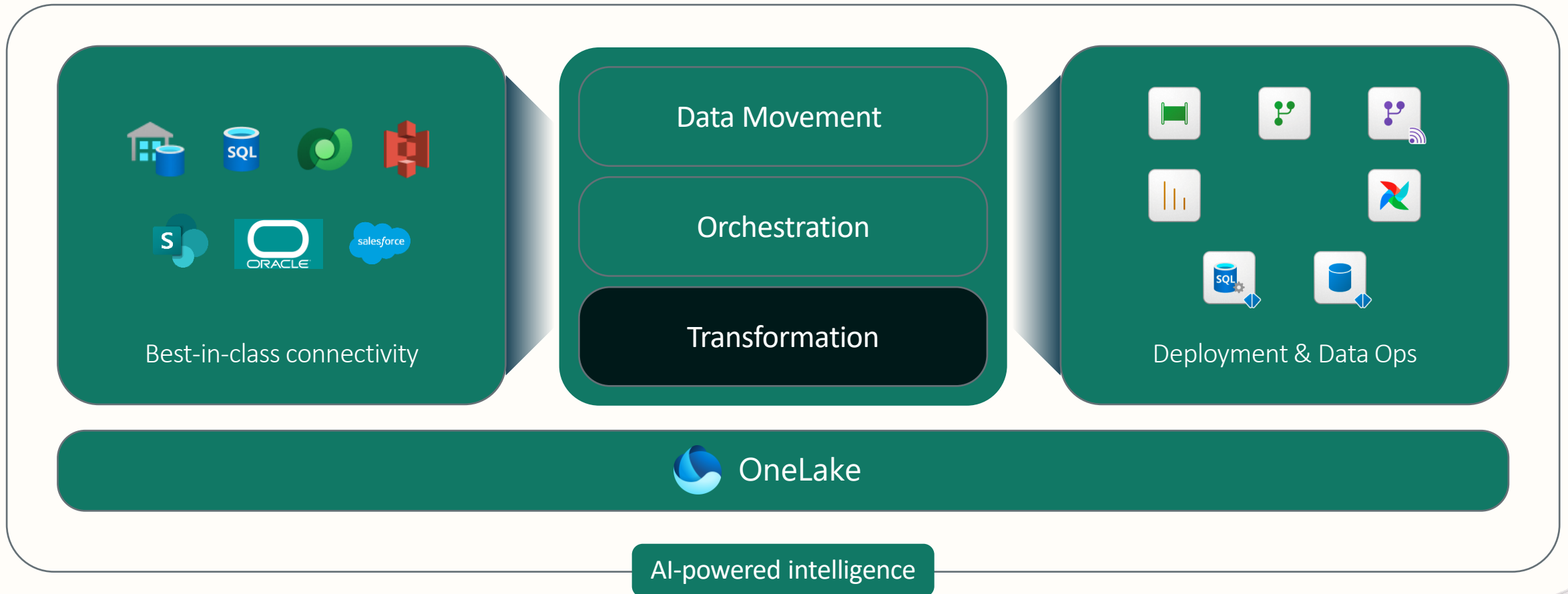
	Name	Type	Opened	Owner	Endorsement	Sensitivity	Workspace
	Data Integration Copilot	Workspace	now	—	—	—	Workspaces
	My workspace	Workspace	5 minutes ago	—	—	—	Workspaces
	Pipeline Copilot	Workspace	44 minutes ago	—	—	—	Workspaces
	Trident-DI-Copilot	Workspace	an hour ago	—	—	—	Workspaces

3

DataFlows Gen2



Data Factory – Data Transformation



Dataflow Gen2

- Cloud-based ETL (Extract, Transform, Load) tool for building and executing scalable data transformation processes
- Experience improvements (validation, diagnostics, etc.)
- More versatility (several Output Destination options)
- High-Scale Data Ingestion & Transformations

The screenshot displays the Microsoft Dataflow Gen2 interface. The top bar shows the Microsoft logo and the title 'Data integration'. Below this is a search bar and a navigation pane on the left. The main workspace is divided into two sections: a query editor and a data table.

The query editor shows a data transformation pipeline. It starts with a 'DimCustomer' table (4 steps) and a 'DimProduct' table (6 steps). These are connected to a 'FactInternetSales' table. The pipeline includes steps for 'Source', 'Navigation', 'Navigation 1', and 'Merged queries'. The 'Merged queries' step is highlighted with a blue box.

The data table at the bottom shows the results of the transformation. It has 33 columns and 99+ rows. The columns are: ProductKey, ProductAlternateKey, EnglishProductName, Color, Size, SizeRange, and Weight. The data is filtered to show rows where the size is not null.

	ProductKey	ProductAlternateKey	EnglishProductName	Color	Size	SizeRange	Weight
1	218	SO-B909-M	Mountain Bike Socks, M	White	4	3	9.50 M
2	219	SO-B909-L	Mountain Bike Socks, L	White	4	3	9.50 L
3	226	LJ-0192-S	Long-Sleeve Logo Jersey, S	Multi	4	3	48.07 S
4	227	LJ-0192-S	Long-Sleeve Logo Jersey, S	Multi	4	3	48.07 S
5	228	LJ-0192-S	Long-Sleeve Logo Jersey, S	Multi	4	3	49.99 S
6	229	LJ-0192-M	Long-Sleeve Logo Jersey, M	Multi	4	3	48.07 M
7	230	LJ-0192-M	Long-Sleeve Logo Jersey, M	Multi	4	3	48.07 M
8	231	LJ-0192-M	Long-Sleeve Logo Jersey, M	Multi	4	3	49.99 M
9	232	LJ-0192-L	Long-Sleeve Logo Jersey, L	Multi	4	3	48.07 L

The bottom status bar indicates 'Completed (2.83 s) Columns: 33 Rows: 99+'. A 'Publish' button is visible in the bottom right corner.

Data Flow Gen2 in Microsoft Fabric

- Power Query meets end users where they work
- Easy data preparation experience
- Embedded in Excel, Power BI, PowerApps and Dynamics 365
- AI powered ETL
- File, relational, multi-dimensional, SaaS data experiences
- Scalable transformations via Fabric compute

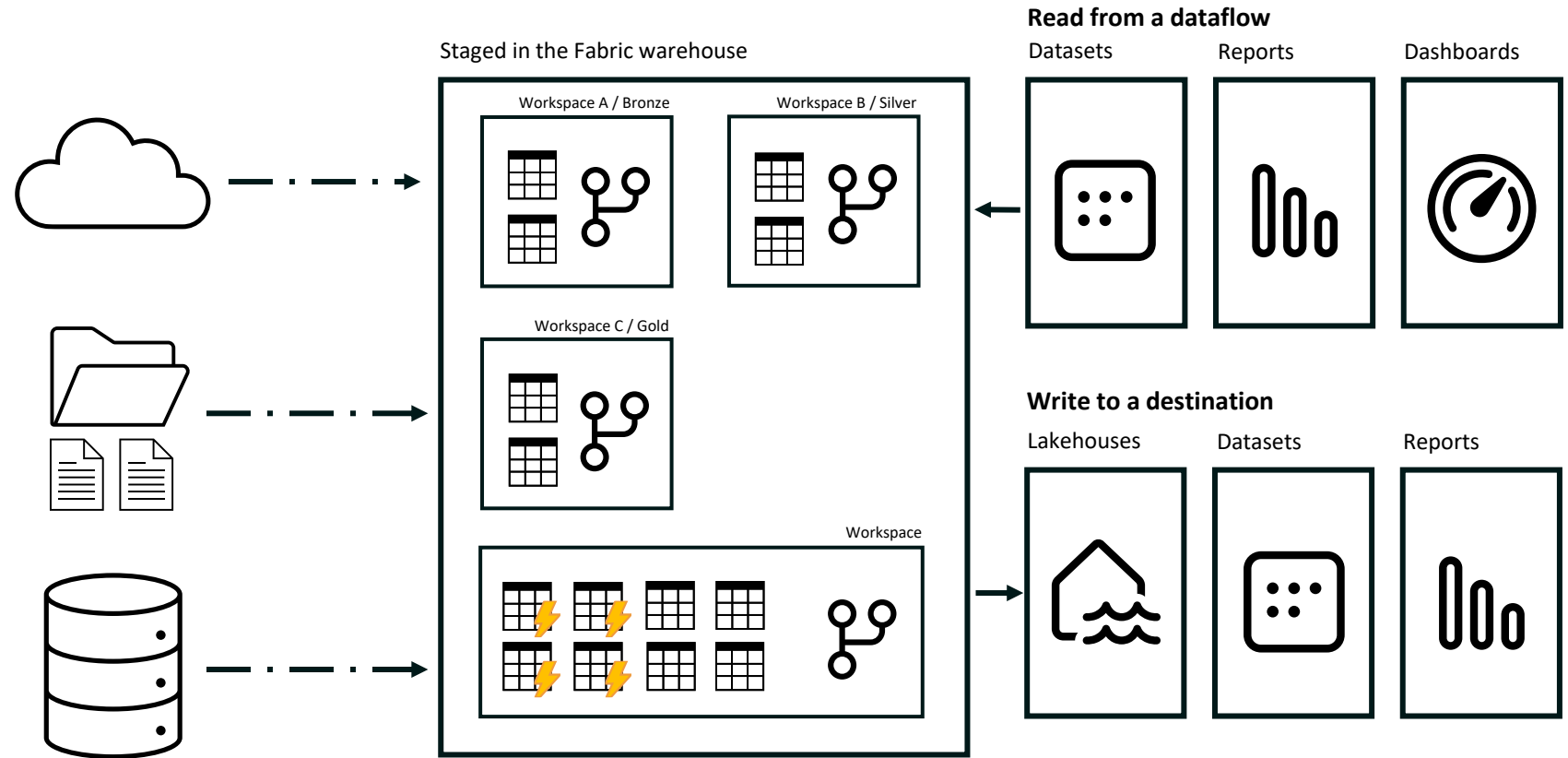
The screenshot displays the Microsoft Excel interface with the Power Query 'Edit queries' pane open. The 'Data' tab is active, and the 'Get Data...' button is highlighted. The Power Query ribbon shows various transformation options like 'Get data', 'Enter data', 'Options', 'Manage parameters', 'Refresh', 'Advanced editor', 'Choose columns', 'Remove columns', 'Keep rows', 'Remove rows', 'Sort', 'Split column', 'Group by', 'Merge queries', 'Append queries', 'Combine files', 'Replace values', and 'Data type'. The 'Queries' pane on the left shows a query named 'Top Customers'. The 'Query settings' pane on the right shows the 'Top Customers' query with a data preview table.

CustomerID	CompanyName	ContactName	ContactTitle	Total Sales
1	QUICK-Stop	Horst Kloss	Accounting Manager	117483.39
2	SAVEA	Jose Pavarotti	Sales Representative	115673.39
3	ERNSH	Roland Mendel	Sales Manager	113236.68
4	HUNGO	Patricia McKenna	Sales Associate	57317.39
5	RATTIC	Paula Wilson	Assistant Sales Representat...	52245.9
6	HANAR	Mario Pontes	Accounting Manager	34101.15
7	FOLKO	Maria Larsson	Owner	32555.55
8	MEREP	Jean Fresnière	Marketing Assistant	32203.9
9	KOENE	Philip Cramer	Sales Associate	31745.75
10	QUEEN	Lúcia Carvalho	Marketing Assistant	30226.1

Dataflow Gen 2

Dataflows are a self-service, cloud-based, data preparation technology.

Dataflows are available in Power BI, Power Apps and Dynamics 365 Customer Insights.

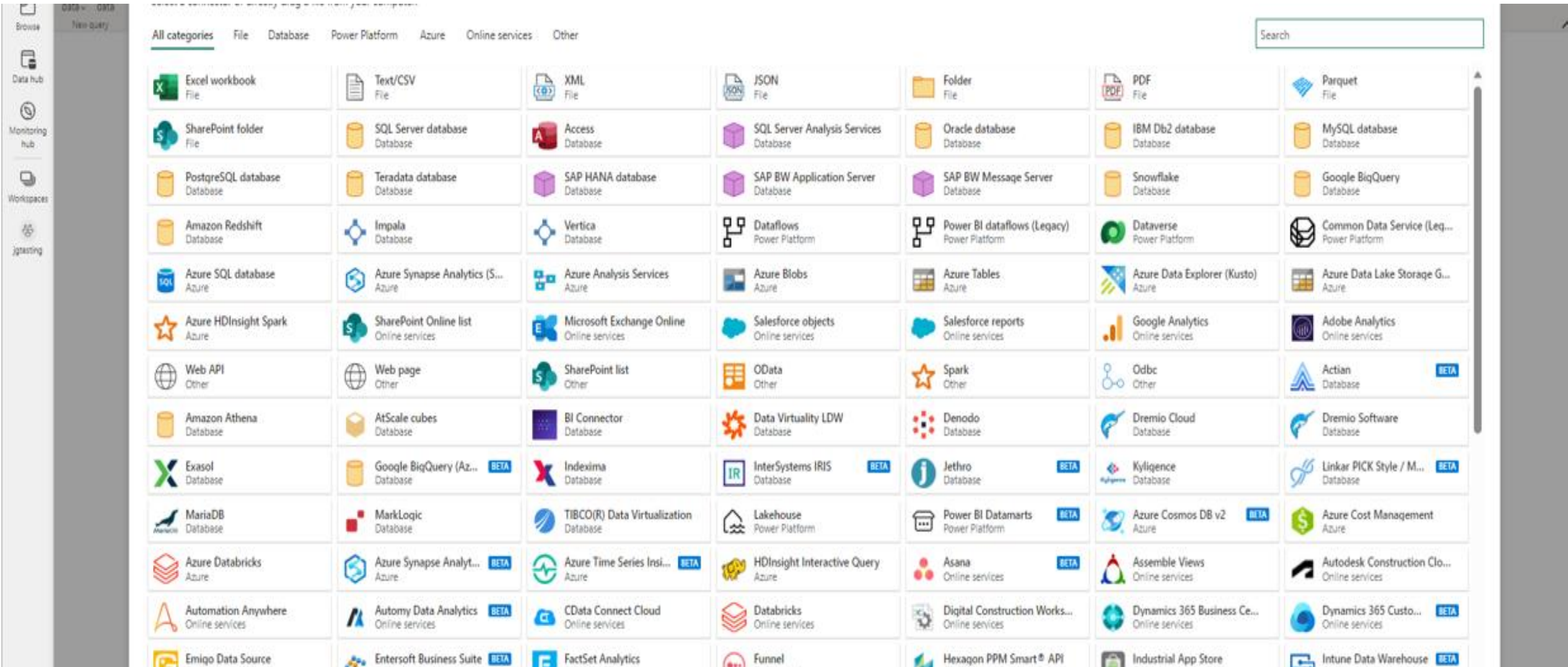


Dataflow Gen1 vs. Dataflow Gen2

 [Differences](#)

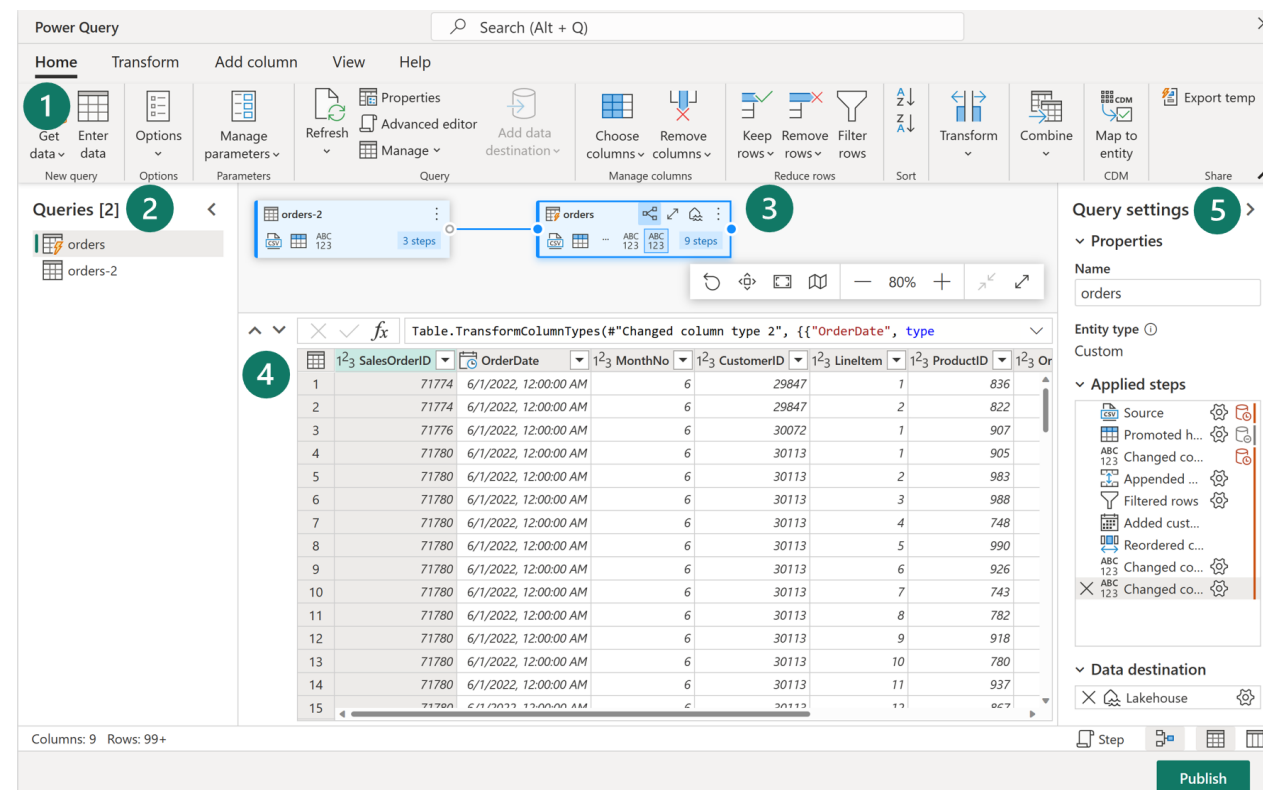
Feature	Dataflow Gen1	Dataflow Gen2
Author dataflows with Power Query	✓	✓
Shorter authoring flow		✓
Auto-Save & background publishing		✓
Data Destinations		✓
Improved monitoring and refresh history		✓
Integration with data pipelines		✓
High-scale compute		✓
Get Data via Dataflows connector	✓	✓
Direct Query via Dataflows connector	✓	
Incremental refresh	✓	
AI Insights support	✓	

Ingest data from 100+ sources with Connectors



Components of a Dataflow

1. Power Query ribbon
 - Data sources, Transformations
2. Queries pane
 - Data sources as queries
3. Diagram view
 - Visually see how the data sources are connected and the different applied transformations
4. Data preview pane
 - Shows a subset of data to allow you to see how transformations affect the data
5. Query Settings pane
 - Includes the Applied Steps
 - Data Destination option to land your data

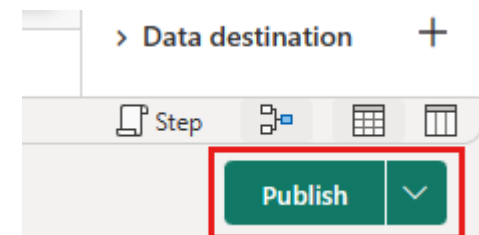
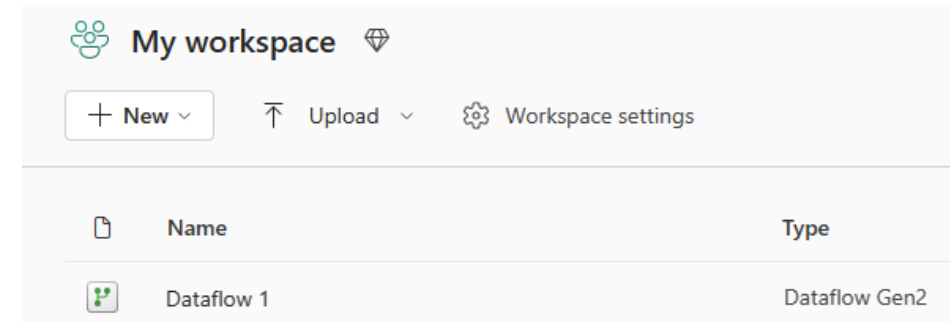
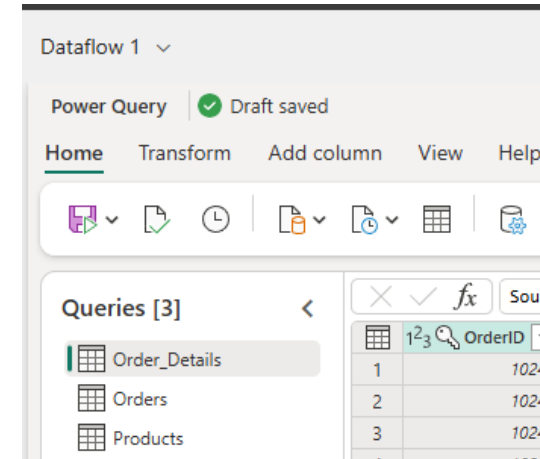


Demo: Create a dataflow

The screenshot displays the Microsoft Data Factory Power Query interface. The top navigation bar includes the Microsoft logo, the text "Data Factory", a search bar, and user profile icons. Below this, the "Power Query" and "Dataflow 11" tabs are visible. The main ribbon contains several groups of tools: "Home" (Get data, Enter data, Options, Manage parameters, Refresh, Properties, Advanced editor, Add data destination), "Transform" (Choose columns, Remove columns, Keep rows, Remove rows, Filter rows, Sort, Split column, Group by, Replace values), "Combine" (Merge queries, Append queries, Combine files), and "Share" (Export template). The central workspace features four large buttons for data import: "Import from Excel" (with an Excel icon), "Import from SQL Server" (with a database cylinder icon), "Import from a Text/CSV file" (with a document icon), and "Import from dataflows" (with a flow diagram icon). Below these buttons are two links: "Get data from another source →" and "Import from a Power Query template". The bottom status bar shows the "Data Factory" logo on the left and a "Publish" button on the right.

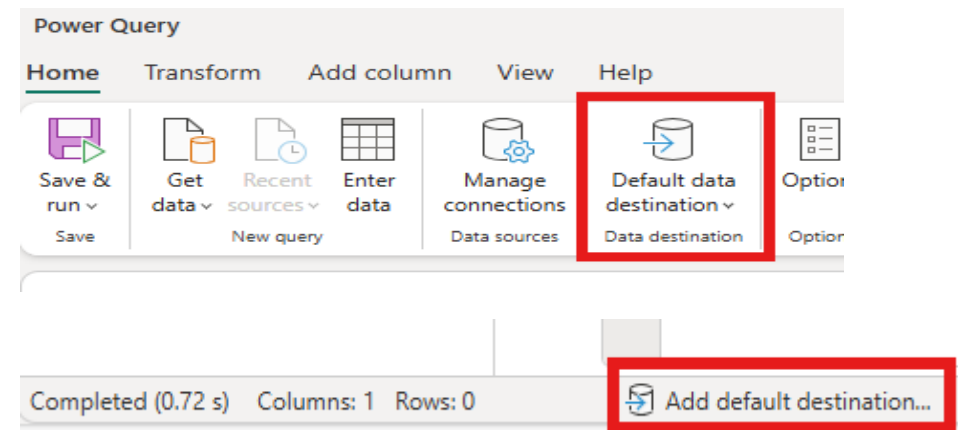
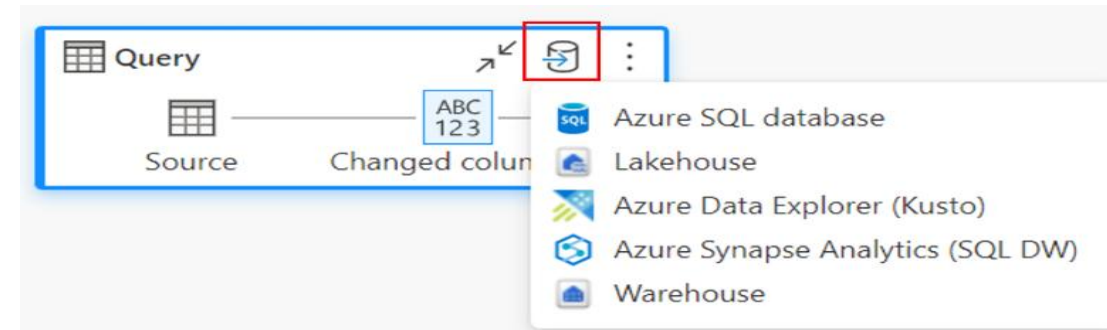
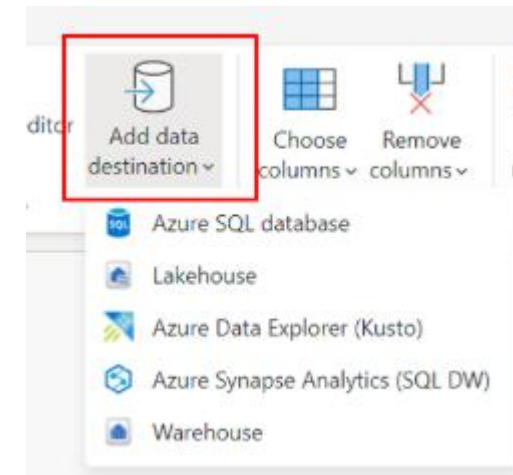
Save a draft of the dataflow

- Draft Version
 - Any change made to a dataflow are auto saved. This saved change is called the draft version
 - This powerful feature allows you to make changes to your dataflow without immediately publishing them to your workspace
 - Helps in not losing changes when session ends abruptly
 - Publishing can be deferred to the last
- Published Version: The version of the dataflow that passed validation and is ready to refresh is called the published version
 - Deferring this operation is useful as it is a long running operation
- To publish:
 - Open the saved dataflow
 - Click Publish on the bottom right corner
 - Dataflow starts the publishing process in the background, indicated by a spinner next to the dataflow name



Dataflow destinations

- To save the prepared data through Dataflows, there are several destinations to choose from
- Choose the destination location and Dataflow writes the data there
- Each query can have a separate destination within a Dataflow
- Ways to setup destination:
 - In the top ribbon – “Add data destination”
 - Query settings – “+ Data destination”
 - Diagram View – Data icon on the top
- Connect to the destination
- Follow through with the rest of the steps (will vary depending on the destination)
- Default Destination: Helps load all queries to the same destination



Staging – how it works

Use staging to improve performance of query processing

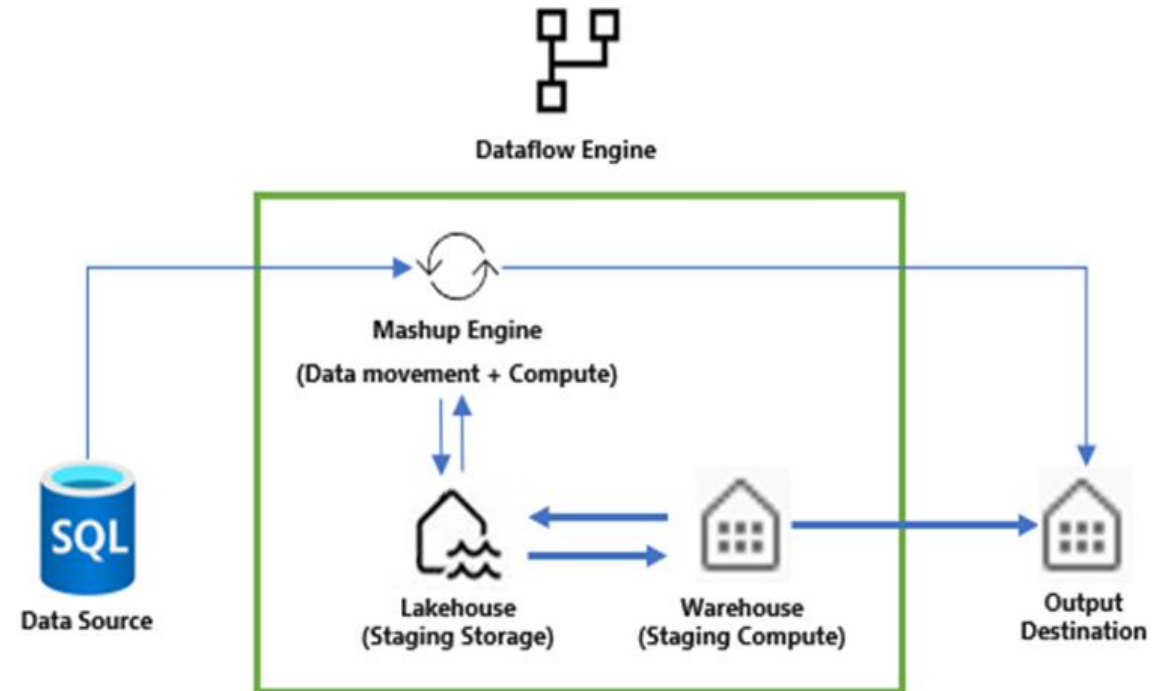
When you create your first Gen2 Dataflow, Staging Lakehouse and Warehouse are provisioned alongside their related SQL Endpoint and Dataset

The Lakehouse is used to stage the data being ingested

The Warehouse used as a compute engine and means to write back results to staging or to supported output destinations faster

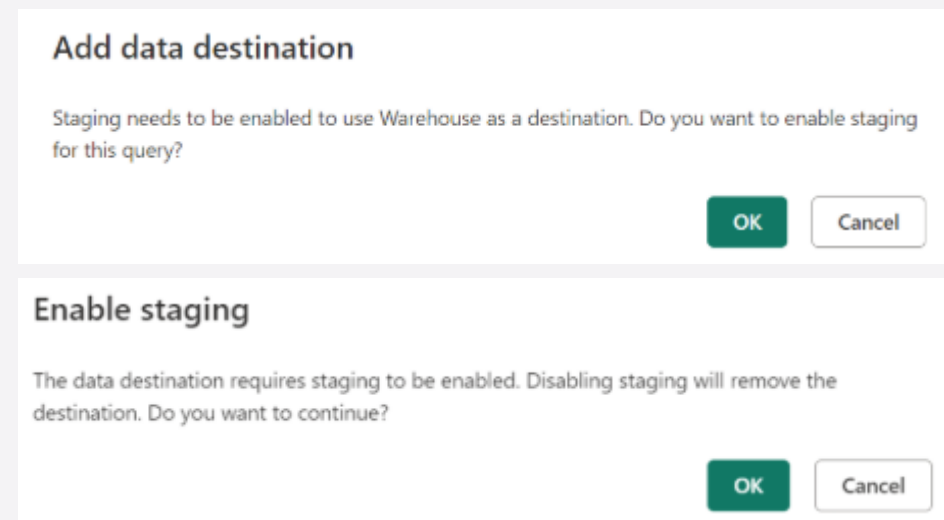
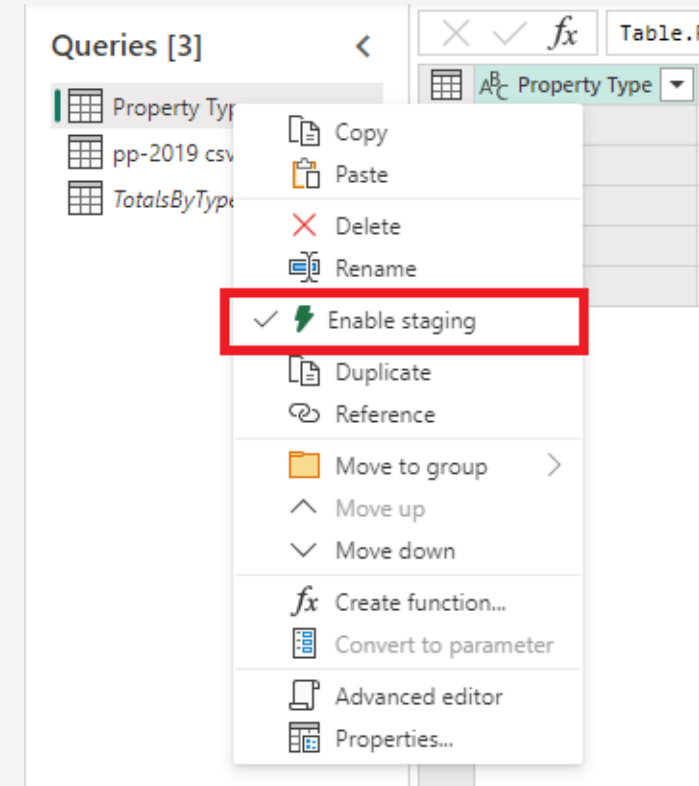
When the warehouse compute can't be used or when staging is disabled, the mashup engine will extract, or load data to staging or destination

Staging is disabled when loading data to lakehouse or non-warehouse destinations



Staging - enabling

- To enable staging of your data – right click on the query name from the Query pane.
- This is **on by default**
- Staging your data could have an impact on performance, so this setting should be tested to ensure the option works for your data
- When staging is disabled, and you choose Warehouse as the output destination, you get a warning
- For a chosen warehouse destination, disabling staging will throw a warning



Staging – Consider this scenario

The screenshot displays the Microsoft Power Query interface within the Data Factory environment. The workspace shows a query named "100KCustomers" with the following applied steps:

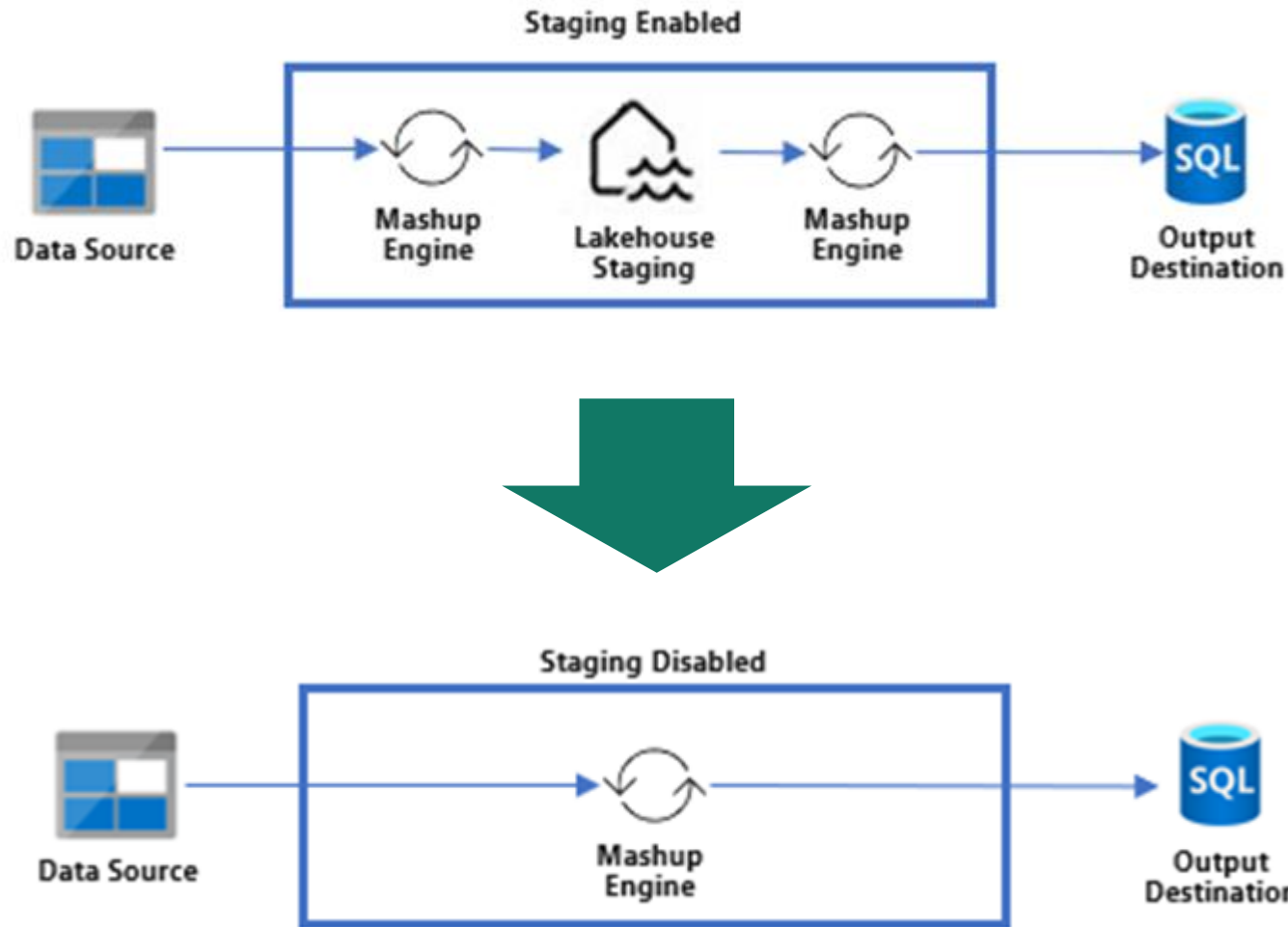
- Source
- Changed col...
- Rename col...
- Removed col...
- Filtered rows

The data preview table at the bottom shows the following columns and rows:

	FirstName	LastName	Full Name	Credit Card Type	CustomerID
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

The status bar at the bottom indicates: Completed (1.19 s) Columns: 5 Rows: 99+.

Staging – When it doesn't make sense



- If your data source does not contain big data volumes
- If you are not joining data from different data sources
- If you are not performing compute/memory intensive transformations such joining or aggregating large data volumes

Fast Copy

To enable fast copy connectors:

- Navigate to Home and select Options
- Within the Global group, select Scale
- Select Allow use of fast copy connectors
- Concurrency
- Dataflows automatically switch to the faster backend when your data size passes a certain threshold
- Bypass the threshold to force fast copy by selecting the Require fast copy setting

Options

Global

- General
- Data load
- Diagnostics

Dataflow

- Data load
- Regional settings
- Privacy
- Scale**
- Parameters

Fast Copy

☒ Allow use of fast copy connectors

Partitioned compute (Preview)

☐ Allow use of partitioned compute

[Learn more](#)

Query Evaluation (Preview)

☐ Allow use of the modern query evaluation engine

[Learn more](#)

Concurrency ⓘ

☐ Limit number of concurrent evaluations

4

OK Cancel

Query properties

Name *

Customers

Description

☐ Enable staging

☒ Require fast copy

OK Cancel

Customers

Employees

Order_Details

Orders

Copy

Paste

Explain this query

Delete

Rename

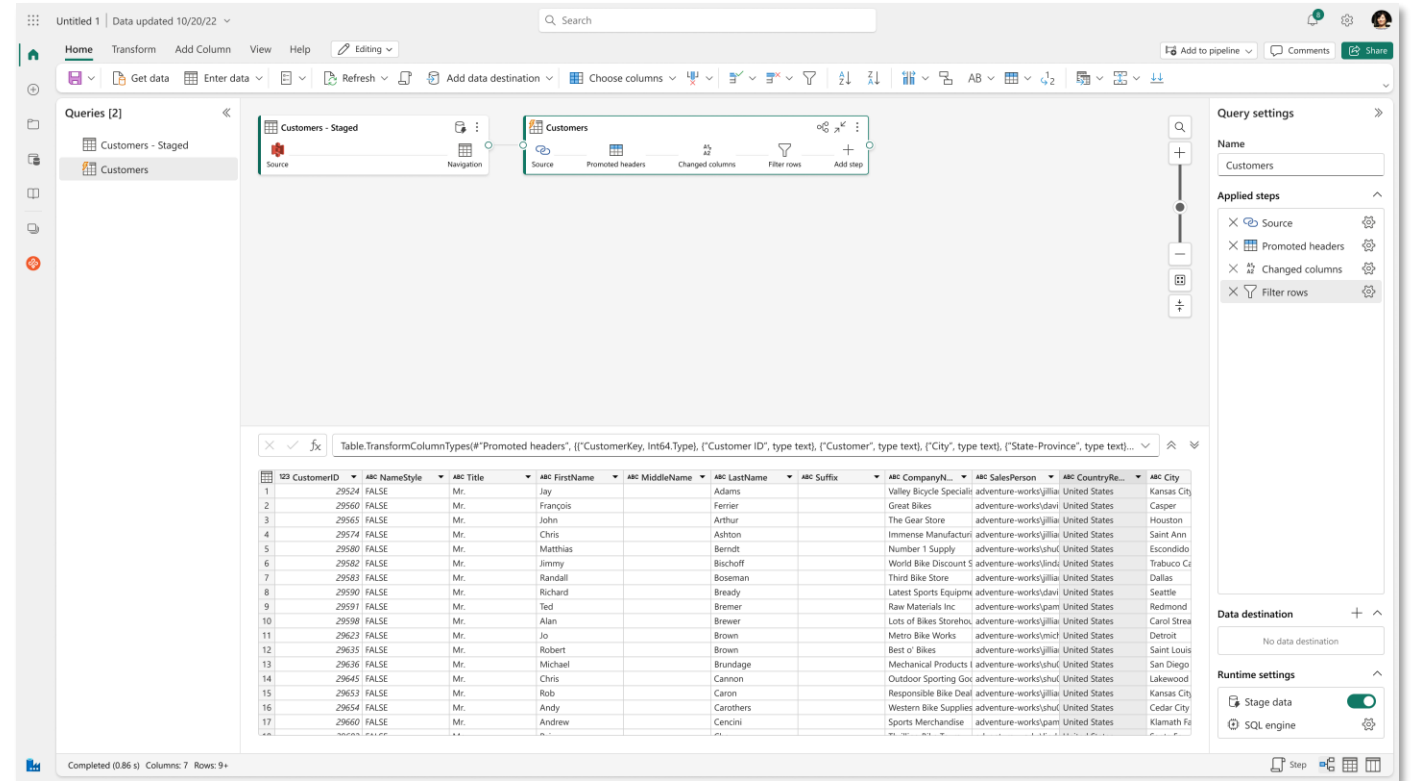
Enable staging

Incremental refresh

Require fast copy

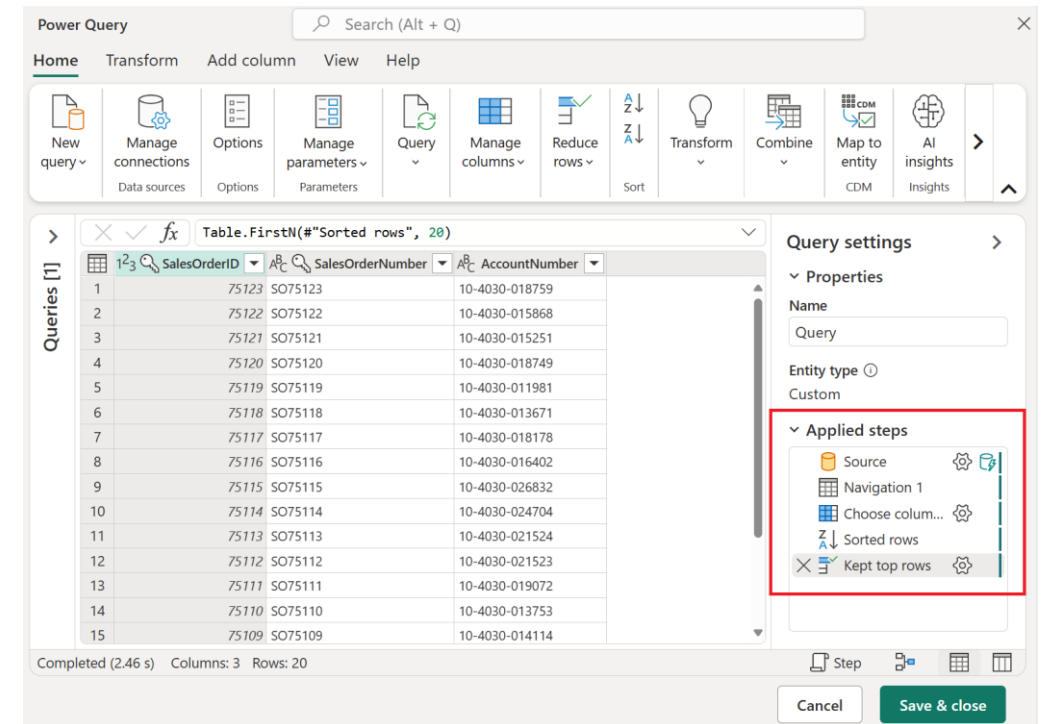
Fast Copy continued

- Get Data with Fast Copy
- Enable the richness of ADF Copy Source/Sink to Fabric Data Factory with a consistent Get Data Experience
- Dataflows Gen2 will enable customers to use fast copy to bring data to a staging area for transformation
- **Sources:** Dataverse, Dynamics 365, Azure Synapse Analytics, Azure Table Storage, SharePoint Online List, Azure Database for PostgreSQL, PostgreSQL, Azure SQL Server, Teradata, Salesforce, Google BigQuery, MongoDB, Sybase and more...

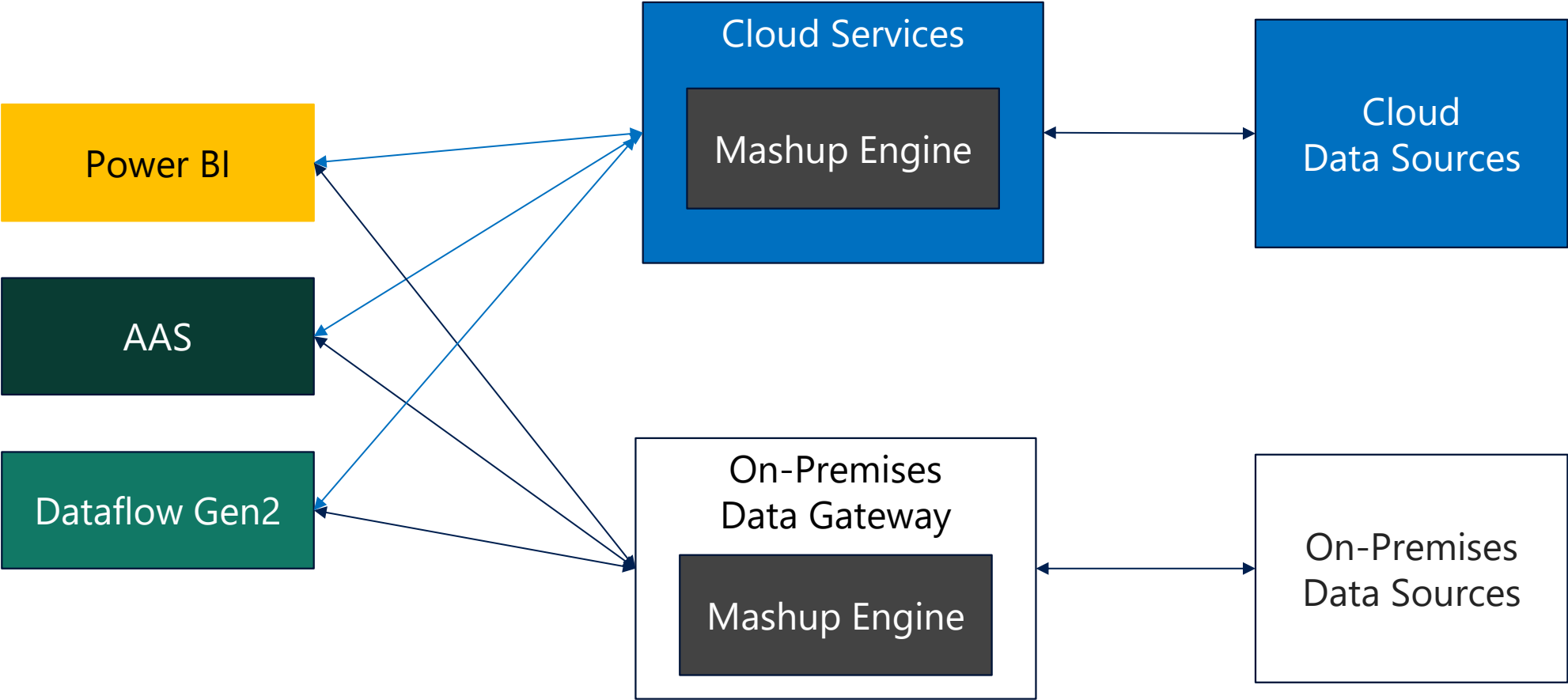


What is the Mashup Engine?

- The Mashup Engine is a platform for data connectivity
- Originally designed as a Self Service ETL product in Excel know as Power Query
- Integrated in the Power BI space to connect to data sources and shape the dataset
- Also used in Dataflows Gen1 and Gen2 to implement Data Transformation Pipelines in the Cloud

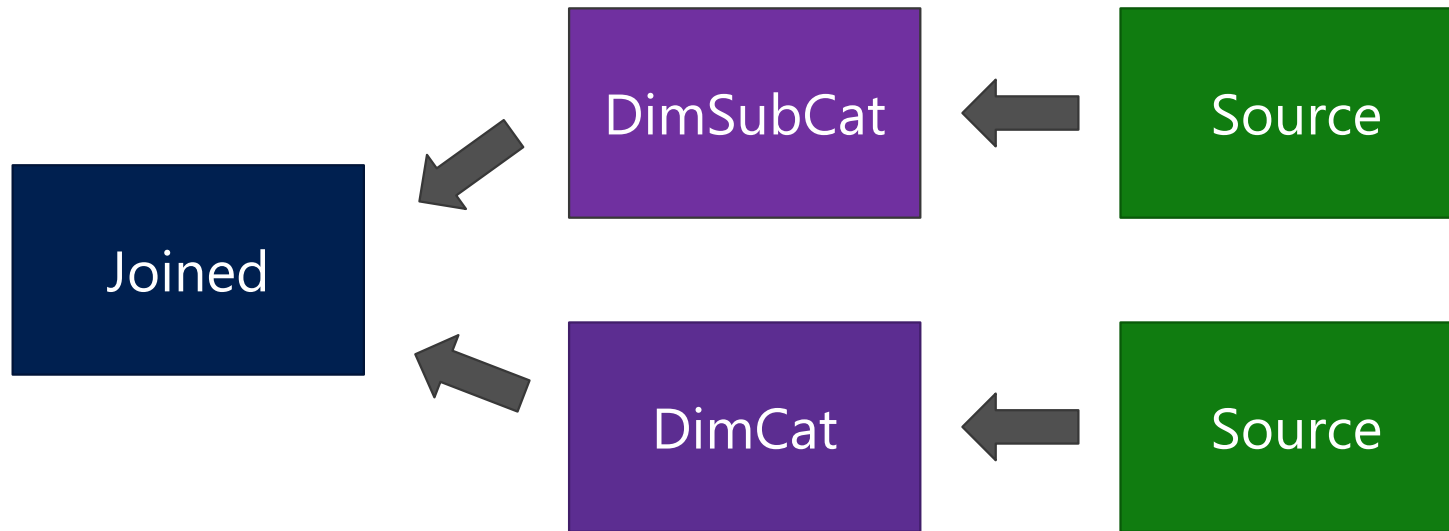


What is the Mashup Engine and Operation Execution

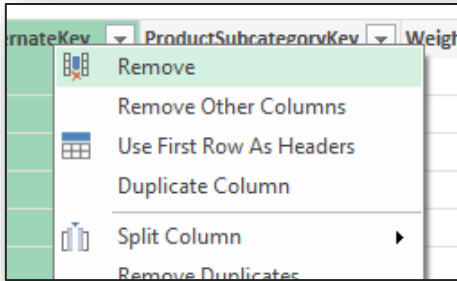


Mashup Queries Evaluation Flow

```
let
  Source = Sql.Database("localhost", "AdventureWorksDW"),
  DimCat = Source{[Schema="dbo",Item="DimProductCategory"]}[Data],
  DimSubCat = Source{[Schema="dbo",Item="DimProductSubcategory"]}[Data],
  DimCustomer = Source{[Schema="dbo",Item="DimCustomer"]}[Data],
  Joined = Table.NestedJoin(DimSubCat,{"CategoryKey"},DimCat,{"CategoryKey"},"Category",JoinKind.Inner)
in
  Joined
```



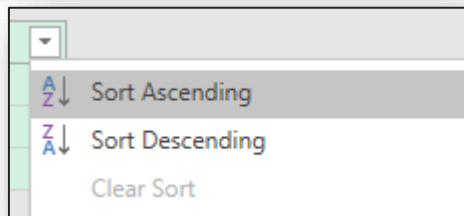
Query Folding (Delegation, Pushdown)



	ProductKey	Product
1		Adjustable Race
2		Bearing Ball
3		BB Ball Bearing
4		Headset Ball Bearings

8	HL Crankarm	
9	Chainring Bolts	
10	Chainring Nut	
11		
12		
13		
14		
15	Decal 2	
16	Down Tube	
17	Mountain End Caps	

A screenshot of a context menu for a table column. The menu is open, showing options: Number Filters, Replace Values..., Drill Down, Add as New Query, Equals, Does Not Equal, Greater Than, Greater Than Or Equal To, Less Than, and Less Than Or Equal To. An orange arrow points from this menu to the 'FilteredRows' line in the code block.

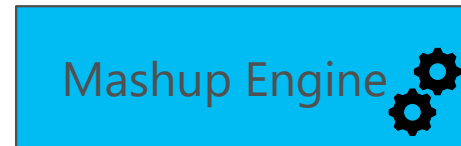


```
let
    Source = Sql.Database("localhost", "AdventureWorksDW2012"),
    Products = Source[Schema="dbo",Item="DimProduct"][Data],
    RemovedOtherColumns =
        Table.SelectColumns(Products,{"ProductKey", "Name"}),
    RenamedColumns =
        Table.RenameColumns(RemovedOtherColumns,{{"Name",
            "Product"}}),
    FilteredRows = Table.SelectRows(RenamedColumns, each
        [ProductKey] < 10),
    Sorted = Table.Sort(FilteredRows,{{"ProductKey",
        Order.Ascending}})
in
    Sorted
```

Folding to Query Languages like SQL

- The Mashup Engine generates queries in the native language of the source, pushing down as much work as possible to the backend. Any remaining work is performed locally.
- Join's across sources can result in filters or join logic being passed from one side to the other.

```
let
  Source = Sql.Database("localhost", "AdventureWorksDW2012"),
  Products = Source{[Schema="dbo",Item="DimProduct"]}[Data],
  RemovedOtherColumns = Table.SelectColumns(Products,{"ProductKey", "Name"}),
  RenamedColumns = Table.RenameColumns(RemovedOtherColumns,{"Name", "Product"}),
  FilteredRows = Table.SelectRows(RenamedColumns, each [ProductKey] < 10),
  Sorted = Table.Sort(FilteredReaders,{"ProductKey", Order.Ascending})
in
  Sorted
```



```
SELECT ProductKey, Name as [Product]
FROM [dbo].[DimProduct]
WHERE [ProductKey] < 10
ORDER BY [ProductKey]
```



Refresh Dataflow Gen2

On demand refresh

Contoso DataPrep

+ New

Upload

Create deployment pipeline

Filter by keyword

Filter

	Name	Type	Owner	Refreshed	Next refresh	Endorsement
☐	IngestCustomersToAzureSQL	Dataflow Gen2	Your User	2/1/24, 3:47:55 PM	N/A	—

Schedule refresh

Refresh

Configure a refresh schedule

Define a data refresh schedule to import data from the data source into the data destination [Learn more](#)

On

Refresh frequency

Daily

Time zone

(UTC) Coordinated Universal Time

Time

1 00 AM

1 00 PM

[Add another time](#)

Send refresh failure notifications to

☒ Dataflow owner

☐ These contacts:

Enter email addresses

Apply Discard

Cancel refresh

Contoso DataPrep

+ New

Upload

Create deployment pipeline

Filter by keyword

Filter

	Name	Type	Owner	Refreshed	Next refresh	Endorsement
☒	IngestCustomersToAzureSQL	Dataflow Gen2	Your User	2/4/24, 4:06:36 PM	N/A	—

Cancel refresh

Edit

Export json

Move to

Properties

Refresh history

Settings

View lineage

Refresh history

Refresh history

IngestCustomersToAzureSQL

Refresh now

Schedule refresh

Download as CSV

Start time	Status	Duration	Type
2/4/2024, 4:09:26 PM	Canceled	00:00:21	On demand

Incremental refresh

- Incremental data extraction from data sources
- Apply Power Query transformations
- Load results to different destinations according to your needs

The screenshot shows the Power Query Desktop interface. On the left, a sidebar contains navigation icons for Home, Workspaces, OneLake, Monitor, Real-Time, Workloads, and Dataflow 2. The main area displays a 'Dataflow 2' with a 'Power Query' tab and a 'Dataflow saved' status. Below the ribbon (Home, Transform, Add column, View, Help), there are buttons for 'Get data', 'Recent sources', 'Enter data', 'Manage connections', 'Default data destination', 'Options', and 'Manage parameters'. A list of queries is shown, including 'Customers' and 'Purchases'. A red circle highlights the three-dot menu icon next to the 'Customers' query. A context menu is open, showing options like 'Expand', 'Highlight related queries', 'Expand related queries', 'Collapse related queries', 'Copy', 'Paste', 'Delete', 'Rename', 'Enable staging', 'Incremental refresh', and 'Require fast copy'. The 'Incremental refresh' option is highlighted with a red bracket. To the right, the 'Incremental refresh' settings dialog is open. It has a title bar with a question mark icon. The first section, 'Incremental refresh', has a checked checkbox for 'Enable incremental refresh'. Below this, there is a prompt 'Choose a DateTime column to filter by *' with a dropdown menu showing 'ReportDate'. The next section, 'Extract data from the past *', has a dropdown for '6' and a unit selector set to 'Years'. The 'Bucket size *' section has a dropdown set to 'Quarter'. The 'Only extract new data when the maximum value in this column changes *' section has a dropdown set to 'ChangeDate'. There is an unchecked checkbox for 'Only extract data for concluded periods'. At the bottom, there is an expandable section for 'Advanced options'. The dialog ends with 'OK' and 'Cancel' buttons.

Incremental refresh ?

☒ Enable incremental refresh

Choose a DateTime column to filter by *

ReportDate

Extract data from the past *

6 Years

Bucket size *

Quarter

Only extract new data when the maximum value in this column changes *

ChangeDate

☐ Only extract data for concluded periods

> Advanced options

OK Cancel

Performance improvements in design time

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

ContosoAnalytics

Fabric

Fabric

RetailGBQ

RetailGBQ

groupByRegions

Confidential\Microsoft Extended

Search

Power Query

Draft saved

Search (Alt + Q)

Home

Transform

Add column

View

Help

Group by

Use first row as headers

Count rows

Transpose

Reverse rows

Count rows

Replace values

Detect data type

Mark as key

Rename

Pivot column

Unpivot columns

Fill

Move

Convert to list

Split column

Format

Parse

Merge columns

Extract

Statistics

Standard Scientific

Number column

Trigonometry

Bounding

Information

Date

Time

Duration

Date and time column

Queries [5]

Transform file fr...

Helper queries

Transform Sample ...

dfd2a0bd-877f-443...

Table.TransformColumnTypes(#"Expanded table column", {{"request_datetime", type datetime}, {"originating_base_num", type

	Source.Name	hvfhs_license_num	dispatching_base_num	originating_base_num	request_datetime
1	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02682	B02682	1/1/2021, 12:45:56 AM
2	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:21:15 AM
3	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:39:12 AM
4	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:46:11 AM
5	NYC_TaxiData/few_csvs_for_one_file/...	HV0005	B02510		1/1/2021, 12:04:00 AM
6	NYC_TaxiData/few_csvs_for_one_file/...	HV0005	B02510		1/1/2021, 12:40:06 AM
7	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:10:36 AM
8	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02875	B02875	1/1/2021, 12:21:17 AM
9	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02875	B02875	1/1/2021, 12:36:57 AM
10	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02875	B02875	1/1/2021, 12:53:31 AM
11	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02835	B02835	1/1/2021, 12:22:58 AM
12	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02835	B02835	1/1/2021, 12:46:44 AM
13	NYC_TaxiData/few_csvs_for_one_file/...	HV0004	B02800		1/1/2021, 12:12:50 AM
14	NYC_TaxiData/few_csvs_for_one_file/...	HV0004	B02800		1/1/2021, 12:35:32 AM
15	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02682	B02682	1/1/2021, 12:10:22 AM
16	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02682	B02682	1/1/2021, 12:25:00 AM
17	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02682	B02682	1/1/2021, 12:44:56 AM
18	NYC_TaxiData/few_csvs_for_one_file/...	HV0005	B02510		1/1/2021, 12:05:04 AM
19	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02836	B02836	1/1/2021, 12:40:44 AM
20	NYC_TaxiData/few_csvs_for_one_file/...	HV0005	B02510		1/1/2021, 12:11:57 AM
21	NYC_TaxiData/few_csvs_for_one_file/...	HV0005	B02510		1/1/2021, 12:30:51 AM
22	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02836	B02836	1/1/2021, 12:34:46 AM
23	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02512	B02512	1/1/2021, 12:10:26 AM
24	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02512	B02512	1/1/2021, 12:24:23 AM
25	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:12:42 AM
26	NYC_TaxiData/few_csvs_for_one_file/...	HV0003	B02764	B02764	1/1/2021, 12:28:07 AM

Query settings

Properties

Name

dfd2a0bd-877f-4435-adac-74d638b...

Applied steps

Source

Filtered hidden files

Added custom

Invoke custom function

Renamed columns

Removed other columns

Removed errors

Expanded table column

Changed column type

Data destination

No data destination

Completed (18.07 s)

Columns: 5

Rows: 99+

Add default destination...

Not run

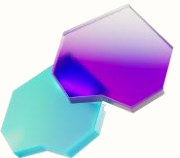
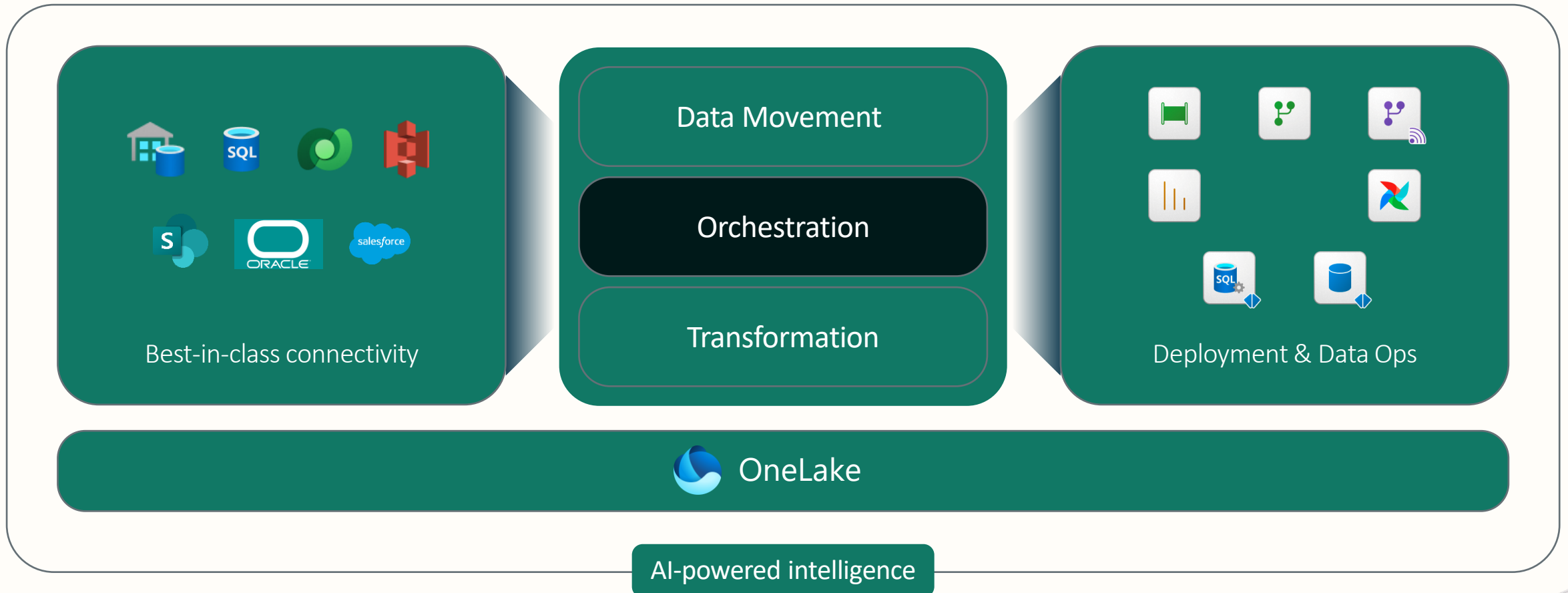
Step

3

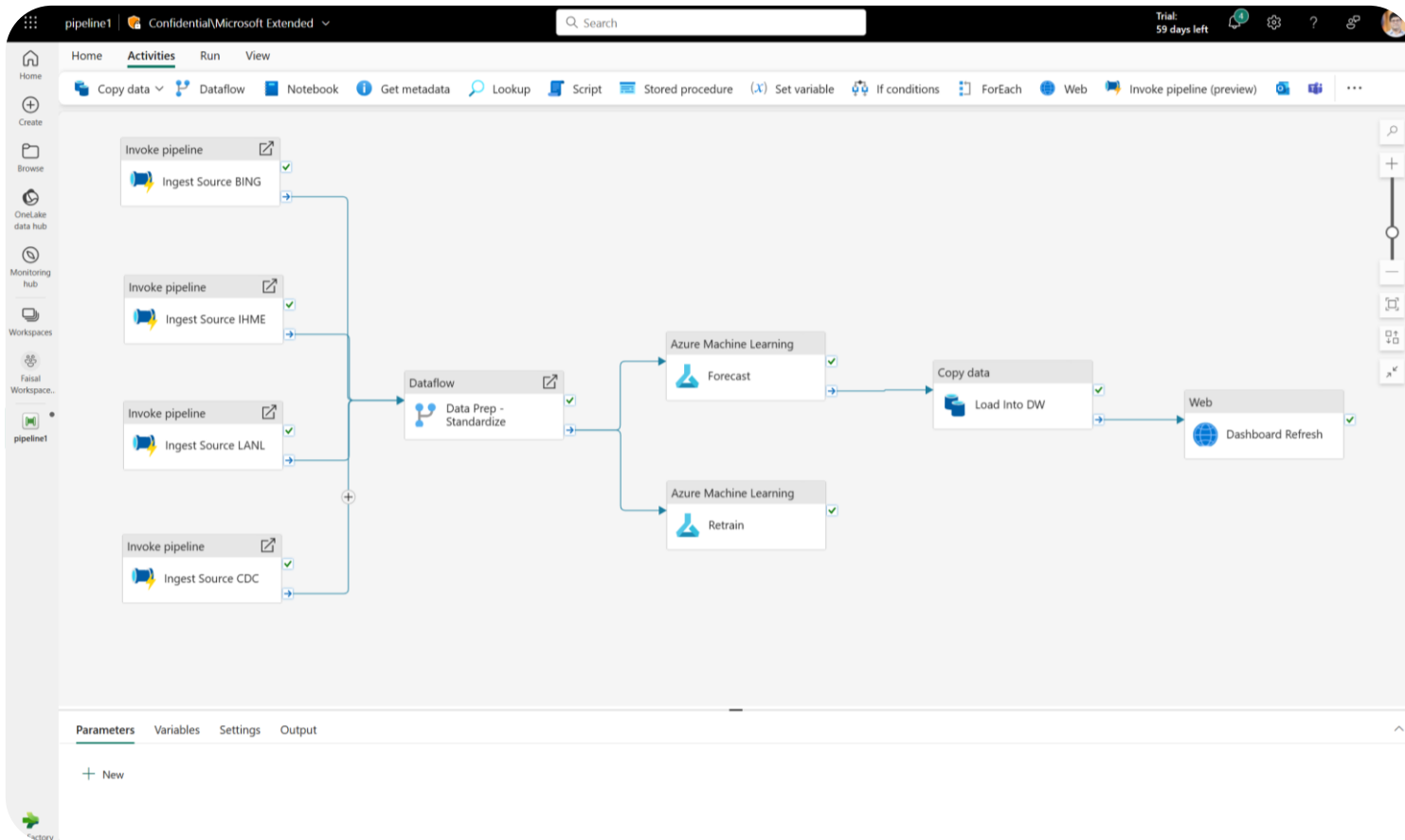
Orchestration



Data Factory - Orchestration



Data Factory Pipelines

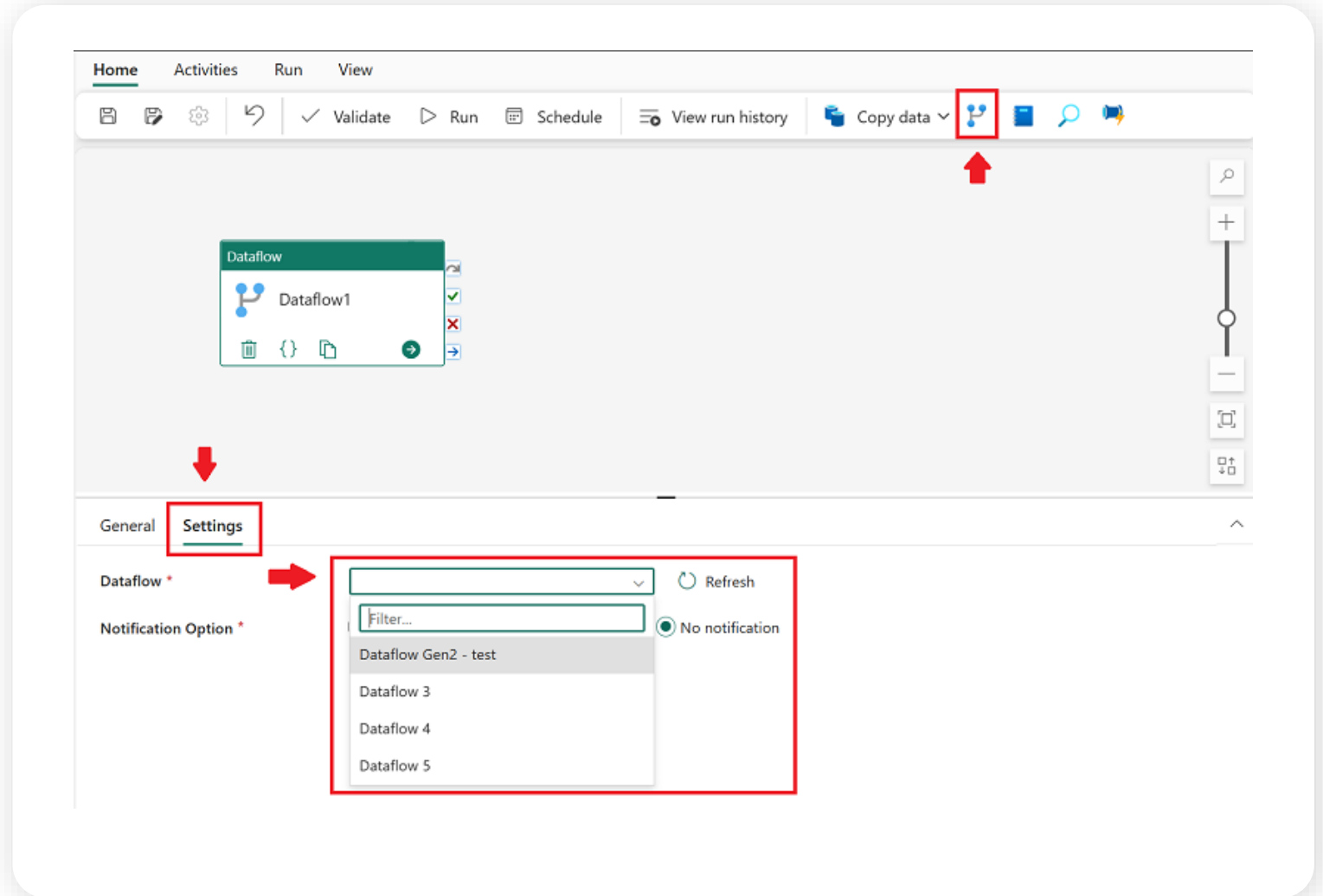


- Intuitive, cloud-based user experience
- Activities to orchestrate data movement, transformation, cleansing, control flow
- Triggered and Scheduled execution models

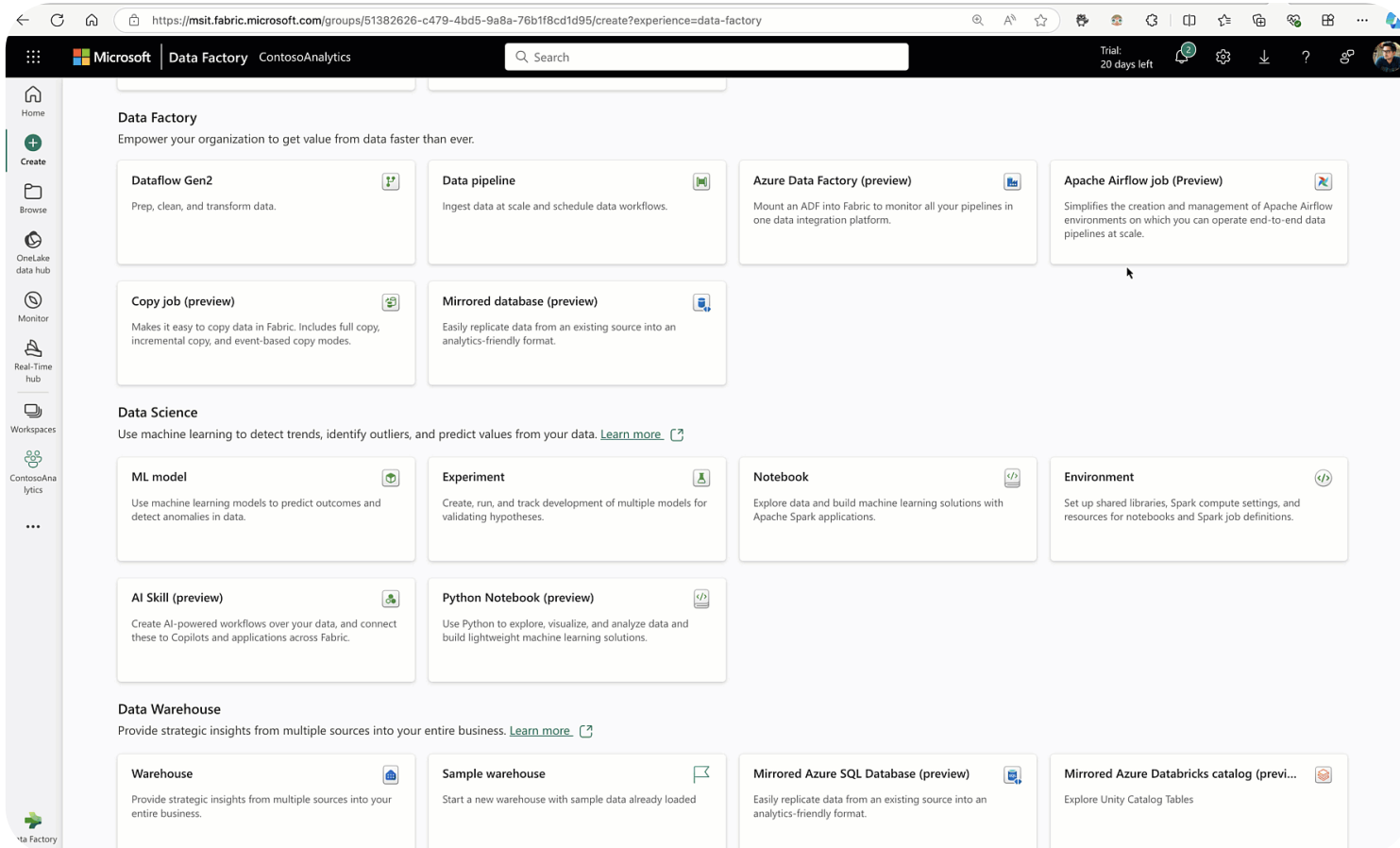
Integrating Dataflows (Gen2) and Pipelines

Use a dataflow for data ingestion and transformation, and landing into a Lakehouse using dataflows

Then incorporate the dataflow into a pipeline to orchestrate extra activities



Apache Airflow in Microsoft Fabric



- Native support for Apache Airflow
- Apache Airflow job – Pro Code, Python based data orchestration based on Open Source Apache Airflow project
- Serverless model with instant provisioning and automatic scaling

Demo

Home

Copilot

Create

Browse

OneLake catalog

Apps

Metrics

Monitor

Learn

Real-Time

Workloads

Workspaces

My workspace

pipeline10

...

Copy data

Copy job (Preview)

Dataflow

Notebook

Get metadata

Lookup

Script

Stored procedure

Set variable

If conditions

...

Try the new pipeline zoom feature for better visibility on large pipelines!

Enable

Copy job (Preview)

Copy job1

Office 365 Email

Office365Email1

Microsoft Teams

MicrosoftTeams1

ForEach

ForEach1

Activities

Notebook1

Dataflow

Dataflow1

Functions

Functions1

Parameters

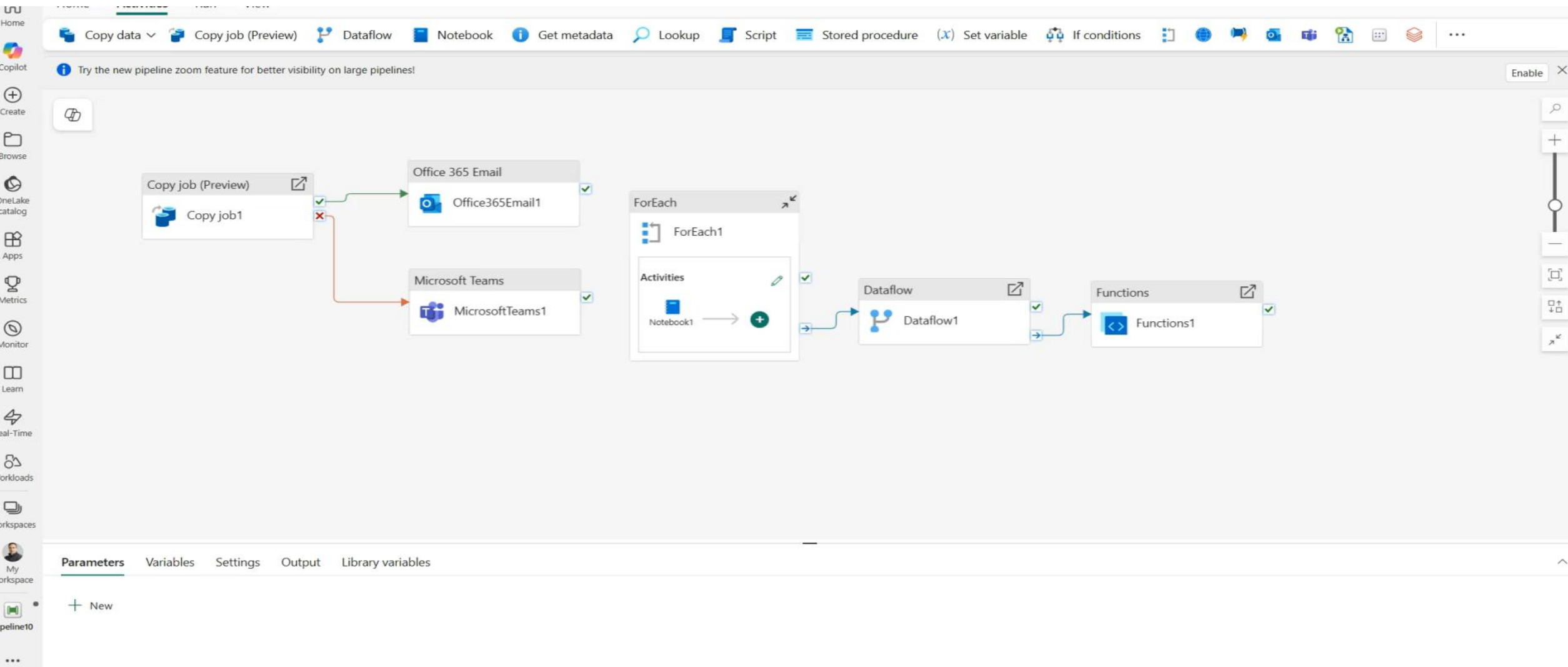
Variables

Settings

Output

Library variables

+ New



Copilot in DataFlows

Easily integrate generative AI into your dataflows and pipelines using Copilot

- Chat with Copilot to describe data transformations in natural language
- Tap into generative AI capabilities from Azure Open AI as data transformation steps
- Use Copilot to schedule and run and manage dataflows

The screenshot displays the Microsoft Fabric Data Factory interface. The main workspace shows a dataflow with two queries: 'Orders - Staging' and 'Orders'. The 'Orders' query is selected, and its details are shown in the bottom pane. The dataflow is composed of several steps: 'Source', 'Split column', 'Open AI', and 'Add step'. The 'Open AI' step is highlighted, and its configuration is shown in the 'Query settings' pane on the right. The 'Query settings' pane includes fields for 'Name' (Orders), 'Applied steps' (Reference, Split column by..., Open AI - Dissatisfact...), and 'Data destination' (No data destination). The 'Copilot' pane on the right shows a chat interface with the prompt 'Create dataflow transforms with Copilot' and a response that describes the data transformation. The bottom pane shows a table with columns: 'Ord', 'City', 'C', 'CustomerReview', 'DissatisfactionReason', 'OrderDate', and 'Due'. The table contains 8 rows of data.

Ord	City	C	CustomerReview	DissatisfactionReason	OrderDate	Due
1	71774	Kansas City	USA	Product was of great quality, but it arrived later than expected. Never got any update from the shipping company as for the reasons for this delay.	6/1/2022, 12:23:01 PM	6/6/2020
2	71776	Casper	USA	Product packaging was damaged. Either a dog tried to eat it or the shipping service didn't treat it with care.	2/24/2023, 08:34:25 PM	3/01/20
3	71780	Houston	USA	This package just showed up on my neighbor's house front porch. I am thankful she knows my name and gave it to me.	1/19/2023, 04:20:19 PM	1/26/20
4	71782	Saint Ann	USA	Product doesn't work - Tried replacing batteries but no luck. Do we have any warranty?	1/29/2022, 07:03:16 PM	2/6/202
5	71783	Escondido	USA	Product never arrived to me, although it says delivery was completed on time.	2/6/2023, 12:00:43 PM	2/13/20
6	71784	Trabuco Canyon	USA	Product is smashed - Need to return it, can someone from your company pick it up at my home address?	3/5/2023, 06:19:09 PM	3/11/20
7	71796	Dallas	USA	I recently purchased a new laptop through your company, and it works well, but battery doesn't last for more than a couple of hours. This is a deal breaker for me as I am usually on the road and need to use without power.	11/30/2022, 04:40:12 PM	12/7/20
8	71797	Seattle	USA	Does your shipping company understand how to provide a good customer service? Very unhappy with the delivery service as the package	1/20/2023, 1:42:32 PM	1/27/20

Home

Transform

Add column

View

Help

Get data

Recent sources

Enter data

Manage connections

Default data destination

Options

Manage parameters

Refresh

Properties

Advanced editor

Add data destination

Choose columns

Remove columns

Reduce rows

Sort

Transform

Combine

Copilot

Export template

Queries [5]

Order_Details

Orders

SalesLT_Customer

Customer_List

Customer

Table.RemoveColumns("#From Value", Table.ColumnsOfType("#From Value", {type table, type record, type list, type nullable

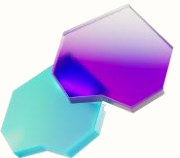
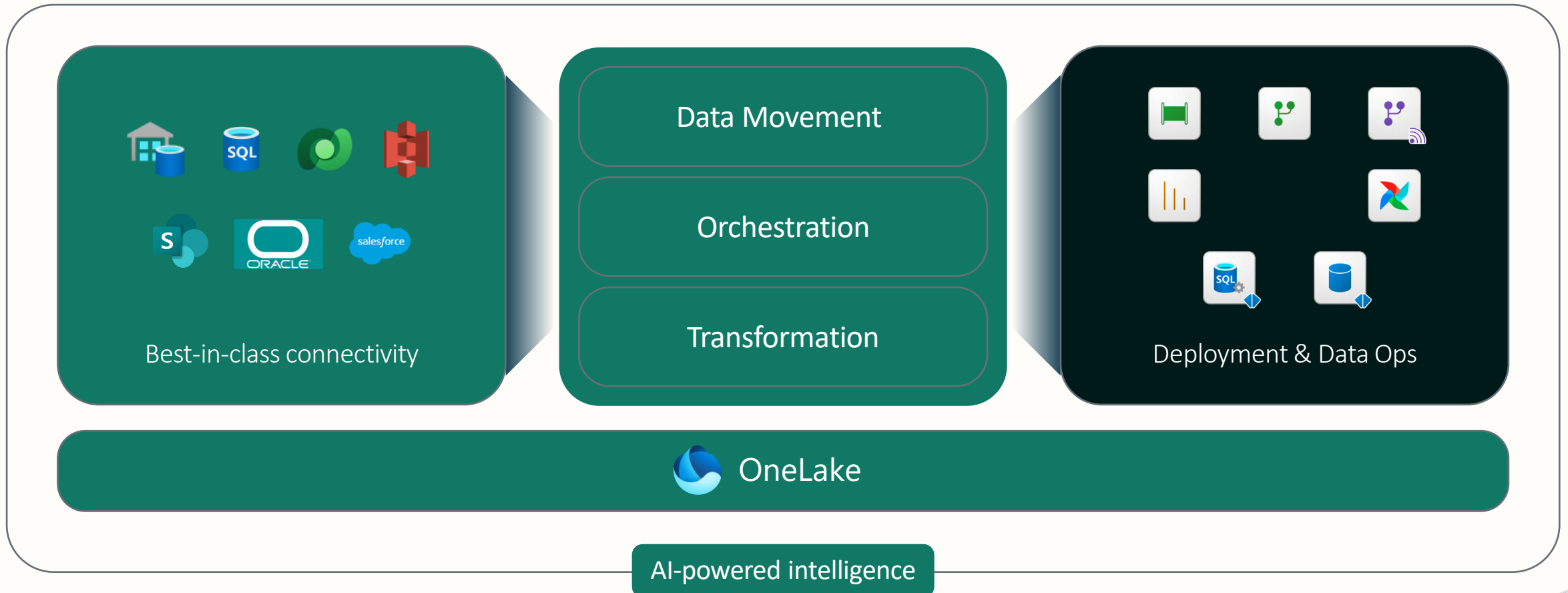
	OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia
1	10248	VINET	5	7/4/1996, 12:00:00 AM +00:00	8/1/1996, 12:00:00 AM +00:00	7/16/1996, 12:00:00 AM +00:00	3
2	10249	TOMSP	6	7/5/1996, 12:00:00 AM +00:00	8/16/1996, 12:00:00 AM +00:00	7/10/1996, 12:00:00 AM +00:00	1
3	10250	HANAR	4	7/8/1996, 12:00:00 AM +00:00	8/5/1996, 12:00:00 AM +00:00	7/12/1996, 12:00:00 AM +00:00	2
4	10251	VICTE	3	7/8/1996, 12:00:00 AM +00:00	8/5/1996, 12:00:00 AM +00:00	7/15/1996, 12:00:00 AM +00:00	1
5	10252	SUPRD	4	7/9/1996, 12:00:00 AM +00:00	8/6/1996, 12:00:00 AM +00:00	7/11/1996, 12:00:00 AM +00:00	2
6	10253	HANAR	3	7/10/1996, 12:00:00 AM +00:00	7/24/1996, 12:00:00 AM +00:00	7/16/1996, 12:00:00 AM +00:00	2
7	10254	CHOPS	5	7/11/1996, 12:00:00 AM +00:00	8/8/1996, 12:00:00 AM +00:00	7/23/1996, 12:00:00 AM +00:00	2
8	10255	RICSU	9	7/12/1996, 12:00:00 AM +00:00	8/9/1996, 12:00:00 AM +00:00	7/15/1996, 12:00:00 AM +00:00	3
9	10256	WELLI	3	7/15/1996, 12:00:00 AM +00:00	8/12/1996, 12:00:00 AM +00:00	7/17/1996, 12:00:00 AM +00:00	2
10	10257	HILAA	4	7/16/1996, 12:00:00 AM +00:00	8/13/1996, 12:00:00 AM +00:00	7/22/1996, 12:00:00 AM +00:00	3
11	10258	ERNSH	1	7/17/1996, 12:00:00 AM +00:00	8/14/1996, 12:00:00 AM +00:00	7/23/1996, 12:00:00 AM +00:00	1
12	10259	CENTC	4	7/18/1996, 12:00:00 AM +00:00	8/15/1996, 12:00:00 AM +00:00	7/25/1996, 12:00:00 AM +00:00	3
13	10260	OTTIK	4	7/19/1996, 12:00:00 AM +00:00	8/16/1996, 12:00:00 AM +00:00	7/29/1996, 12:00:00 AM +00:00	1
14	10261	QUEDE	4	7/19/1996, 12:00:00 AM +00:00	8/16/1996, 12:00:00 AM +00:00	7/30/1996, 12:00:00 AM +00:00	2
15	10262	RATTC	8	7/22/1996, 12:00:00 AM +00:00	8/19/1996, 12:00:00 AM +00:00	7/25/1996, 12:00:00 AM +00:00	3
16	10263	ERNSH	9	7/23/1996, 12:00:00 AM +00:00	8/20/1996, 12:00:00 AM +00:00	7/31/1996, 12:00:00 AM +00:00	3
17	10264	FOLKO	6	7/24/1996, 12:00:00 AM +00:00	8/21/1996, 12:00:00 AM +00:00	8/23/1996, 12:00:00 AM +00:00	3
18	10265	BLONP	2	7/25/1996, 12:00:00 AM +00:00	8/22/1996, 12:00:00 AM +00:00	8/12/1996, 12:00:00 AM +00:00	1
19	10266	WARTH	3	7/26/1996, 12:00:00 AM +00:00	9/6/1996, 12:00:00 AM +00:00	7/31/1996, 12:00:00 AM +00:00	3
20	10267	FRANK	4	7/29/1996, 12:00:00 AM +00:00	8/26/1996, 12:00:00 AM +00:00	8/6/1996, 12:00:00 AM +00:00	1
21	10268	GRONR	8	7/30/1996, 12:00:00 AM +00:00	8/27/1996, 12:00:00 AM +00:00	8/12/1996, 12:00:00 AM +00:00	3

4

Miscellaneous



Data Factory – Deployment & Data Ops



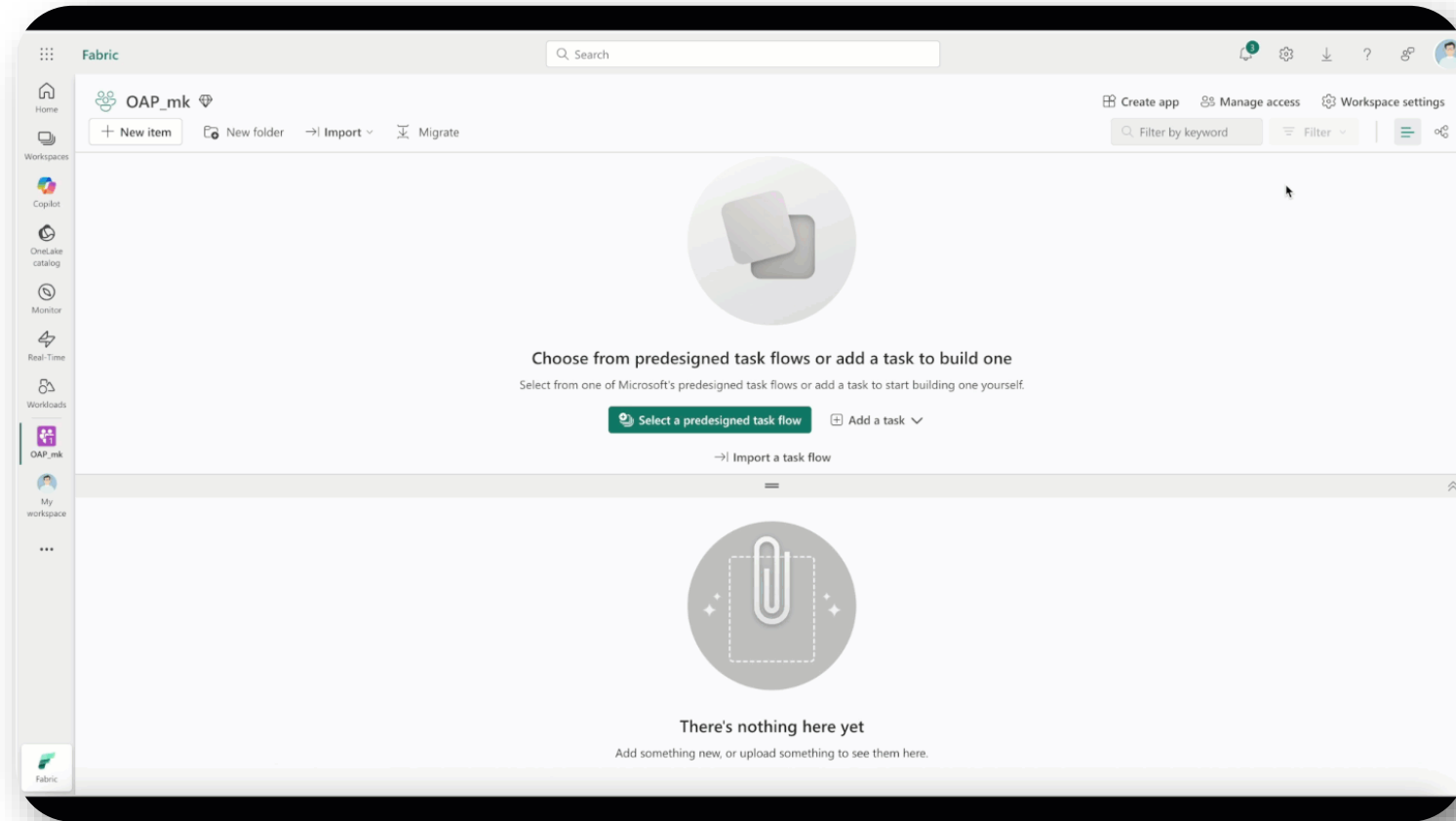
Data Factory – Monitoring, DataOps

The screenshot displays the Microsoft Data Factory Monitoring hub. The main view is a table of activities with columns for Activity name, Status, Item type, Start time, Submitter, and Location. The table lists various pipelines and notebooks, some succeeded and some failed. A 'Workspace settings' dialog is open, showing the 'Git integration' tab. This dialog includes fields for Organization (sbahadur0592), Project (MS-Sales), Git repository (saledev), and Branch (contoso-dev), along with a 'Git folder' field. The 'Connect and sync' button is highlighted.

Activity name	Status	Item type	Start time	Submitter	Location
demo-pipeline	Succeeded	Data pipeline	11:46 PM, 4/26/23	Shireen Bahadur	ContosoAnalytics
demo-pipeline	Succeeded	Data pipeline	11:48 PM, 4/26/23		
Contoso_Data_Pipeline	Failed	Data pipeline	5:28 PM, 5/9/23		
Contoso_Data_Pipeline	Failed	Data pipeline	5:31 PM, 5/9/23		
pipeline6	Succeeded	Data pipeline	10:31 AM, 5/10/23		
Contoso_Data_Pipeline	Failed	Data pipeline	1:36 PM, 5/16/23		
pipeline8	Failed	Data pipeline	11:34 AM, 5/31/23		
Notebook 2_990e4f3d-8fb0-468e-9b4c-844462...	Succeeded	Notebook	11:36 AM, 5/31/23		
pipeline8	Failed	Data pipeline	8:53 AM, 6/2/23		
pipeline10	Succeeded	Data pipeline	11:54 PM, 6/18/23		
pipeline10	Succeeded	Data pipeline	11:54 PM, 6/18/23		
pipeline10	Succeeded	Data pipeline	1:54 PM, 6/19/23		

- Rich monitoring capabilities and complete DataOps
- Connect to favorite Git repositories
- Diff/merge data pipelines
- View and track all your activities across all your workspaces
- CI/CD is now available for all Data Factory items including Airflow, making team collaboration and development seamless
- API for Connections and Gateways are Generally Available
- PowerShell cmdlets for Gateway automation supports takeover/ moving gateway across machines

Data Factory – Outbound access control



Mission Critical capabilities

- Credential-free authentication using Workspace Identity
- Granular network security control for accessing Fabric workspaces (including Data Factory items) with Workspace Private Link
- Protect against Data Exfiltration risks with Workspace Outbound Access Control

New innovation

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

OAP_Test

My workspace

...

OAP_Test

+ New item

New folder

→ Import

Migrate

Create deployment pipeline

Create app

Manage access

Workspace settings

Filter by keyword

Filter

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

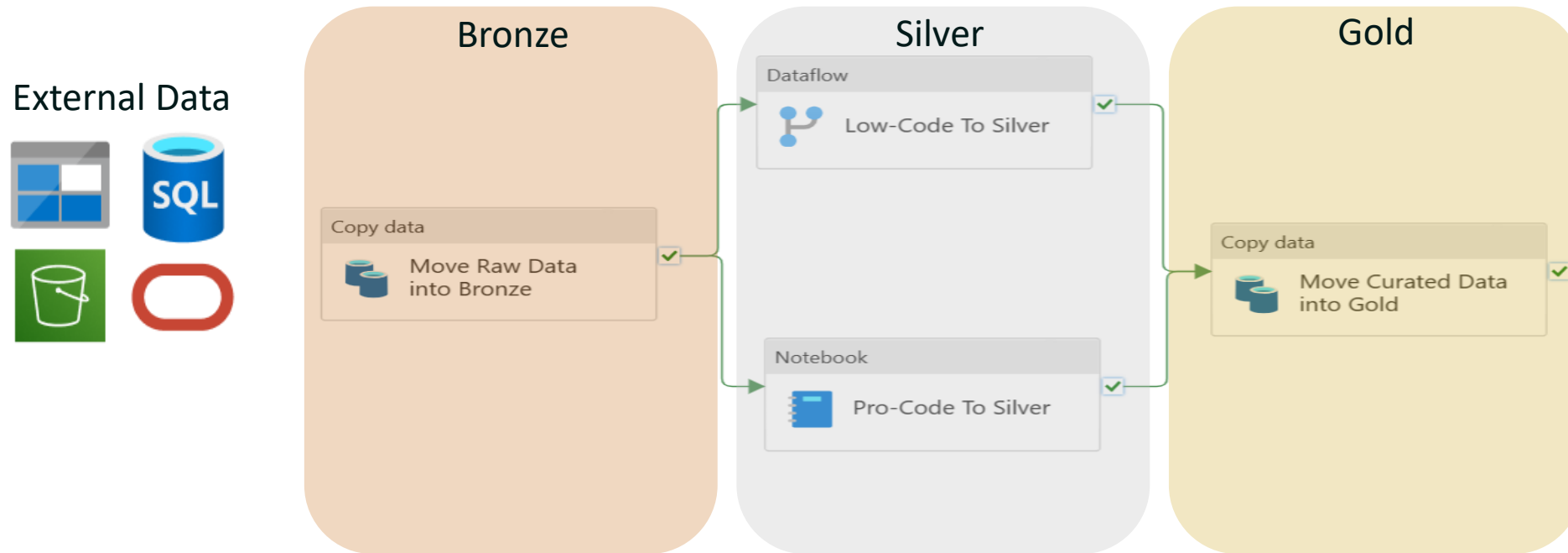
Select a predefined task flow

Add a task

→ Import a task flow

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsemen	Sensitivity	Included in app
pipeline_example		Pipeline	—	GlobalAdmin	—	—	—	—	

Medallion Architecture with Fabric Data Factory

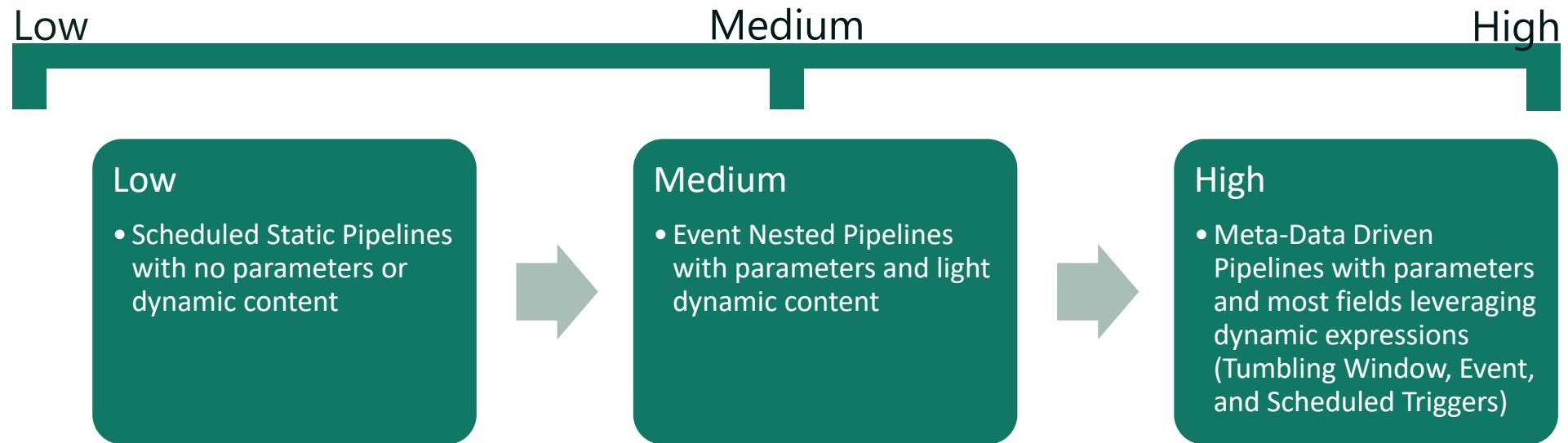


In a minimalist sense, the Medallion Architecture can be actualized using Pipelines to issue a Copy Activity to move external Data into the Fabric OneLake, then immediately upon success, invoke a Transformation Activity, such as Dataflows for low-code or Notebooks for pro-code to source from Bronze, apply transformations, then land into the Silver layer.

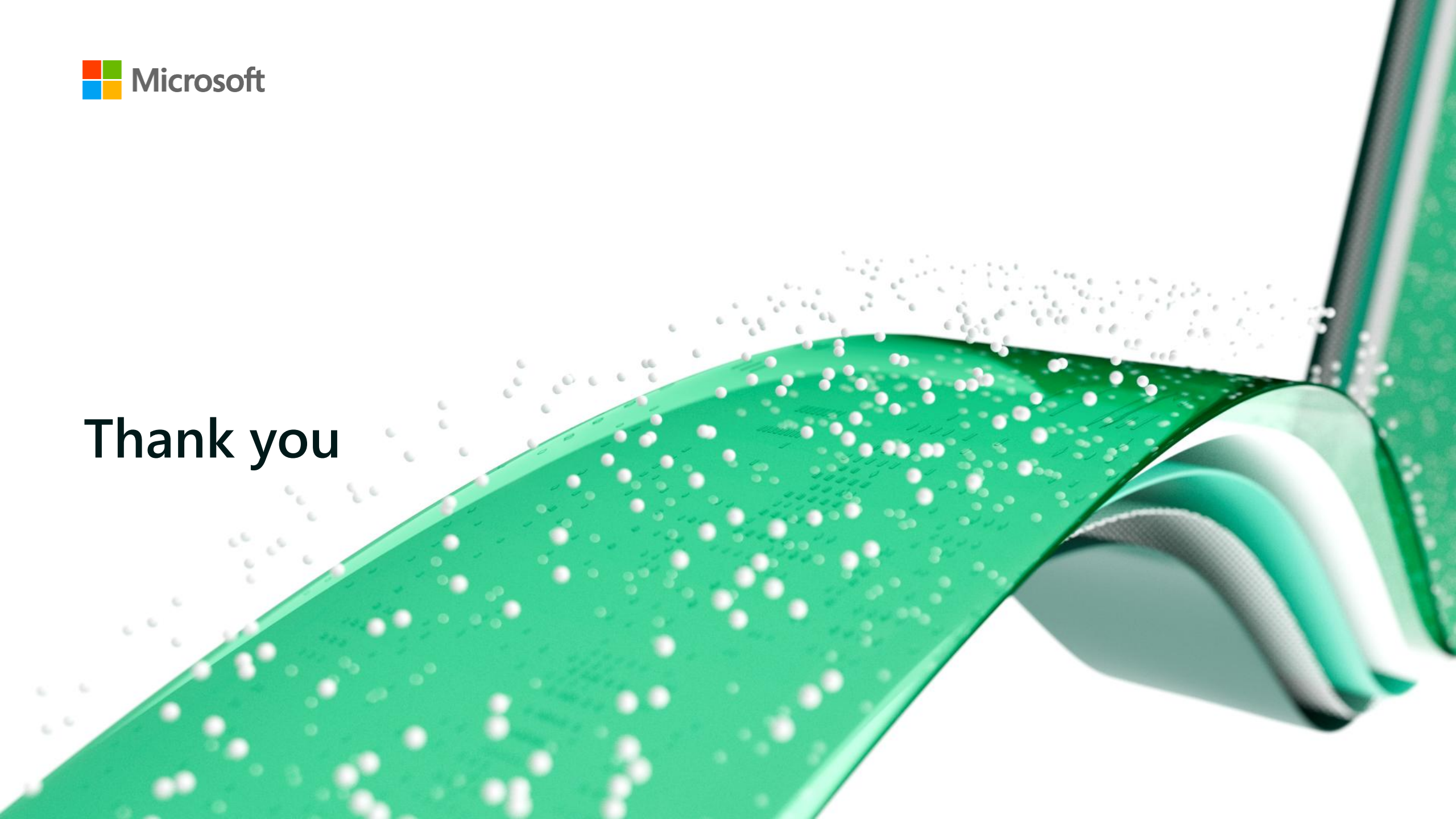
To get to gold, the user can then issue another Copy Data Activity to push this curated data to wherever they need.

Architecture Design Patterns

Given the vastness of features, architecture designs of Data Pipelines often vary across a wide spectrum of complexity



Thank you

An abstract graphic featuring a vibrant green, curved, ribbon-like shape that flows from the bottom left towards the top right. The surface of this green shape is covered with numerous small, white, spherical particles, some of which are also scattered in the air above the ribbon. The background is a clean, solid white.